Does Agricultural Aid Raise Farm Productivity? Evidence from West Africa

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Abstract

Does foreign aid raise productivity? Cross-country evidence on aid effectiveness is decidedly mixed: aggregate aid shows no robust growth effect, while productive, early-impact flows appear more promising, a tension that country-level data cannot resolve because it cannot isolate the channel through which any one aid flow operates. We resolve it at the household level, linking district-level World Bank agricultural aid disbursements to nearly 29,000 household-year observations from the 2018 and 2021 Harmonized Survey on Household Living Conditions across eight West African (WAEMU) countries, and instrumenting district-level disbursements with the average disbursements received by other districts in the same country-year. A 10 percent increase in agricultural aid disbursements raises crop yields by about 1.3 percent. The gains extend to net agricultural income and labor productivity, are strongest for cereals, and are shared across the productivity distribution. They are substantially larger for rural and less-educated households. Higher disbursements improve access to financing and agricultural information, increase the use of purchased inputs, promote mechanization, and encourage crop diversification. These findings provide causal evidence that agricultural aid raises farm productivity and strengthens rural livelihoods in West Africa.

Keywords: Foreign aid; agricultural yield; World Bank; West Africa; instrumental variables; household panel data; leave-one-out instrument; quantile regression.
JEL classification codes: F35; O13; Q12; O55; C26.

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1 Introduction

Smallholder agriculture remains central to livelihoods across Sub-Saharan Africa, yet agricultural productivity continues to lag behind its potential. In West Africa, more than 60 percent of the population depends on farming for income, and cereal yields in the Sahel are roughly half those observed in South Asia despite broadly comparable agroecological conditions (African Development Bank and Food and Agriculture Organization, 2015; World Bank, 2008). Closing this productivity gap requires sustained investment in rural infrastructure, input systems, and technology diffusion. The World Bank is among the largest external financiers of such investment in the region, committing billions of dollars to agricultural projects across West Africa over the past two decades. Whether these resources translate into measurable gains at the farm level, however, remains poorly understood. This paper addresses this question by estimating the causal effect of World Bank agricultural disbursements on household farm productivity in West Africa.

The macroeconomic literature on foreign aid effectiveness has produced mixed evidence. Studies operating at the country level struggle to isolate the productivity mechanisms through which aid may improve household welfare, and cross-sectional designs cannot rule out that more aid simply flows to districts with the weakest underlying conditions (Burnside and Dollar, 2000; Rajan and Subramanian, 2008). Studies that address aid's endogenous allocation more directly often report more positive effects (Dreher and Langlotz, 2020; Clemens et al., 2012), highlighting the importance of identification in assessing aid effectiveness. Existing micro-level evidence focuses primarily on food aid (Nunn and Qian, 2014). Evidence on the productivity effects of agricultural project financing remains limited, in part because estimating such effects requires household-level agricultural data that can be linked to geographically disaggregated aid flows while adequately addressing the endogenous placement of projects. As a result, evidence on the causal effect of agricultural project financing on household farm productivity remains limited.

This paper addresses that gap. We estimate the causal effect of World Bank agricultural disbursements on farm yields using a harmonized panel of approximately 28,700 household-year observations drawn from nationally representative agricultural surveys across eight West African countries (Benin, Burkina Faso, Côte d'Ivoire, Guinea-Bissau, Mali, Niger, Senegal, and Togo). Combining these surveys with geo-coded World Bank project data disaggregated at the second administrative level (ADM2), we exploit variation in district-level aid exposure across the two survey rounds, 2018 and 2021.

The central empirical challenge is endogeneity: aid is not randomly assigned. Districts with persistently weak agricultural conditions may attract more resources, biasing naive estimates downward. We address this with a leave-one-out (LOO) instrument: for each district and year, local disbursements are instrumented with the average disbursements of all other districts in the same country-year. This strategy isolates variation driven by national budget cycles and World Bank portfolio decisions rather than local agricultural conditions, while household fixed effects absorb all time-invariant confounders at the farm level.

Our main finding is clear: World Bank agricultural disbursements significantly raise household farm yields. The preferred 2SLS estimate implies a yield semi-elasticity of 0.127: a 10 percent increase in agricultural disbursements raises yields by approximately 1.3 percent ($p < 0.01$, Kleibergen–Paap $F = 18.49$). This finding helps reconcile apparently conflicting evidence in the aid literature. Rather than examining heterogeneous aid flows at the country level, we isolate a well-defined category of productive investment and identify its effect using household-level exposure to district-level disbursements. This design reveals a farm-level productivity channel that aggregate cross-country analyses may lack the resolution to detect.

Several additional results clarify the nature of this effect. The estimated relationship is robust to alternative measures of aid, including active project counts and project presence indicators, and remains unchanged when controlling for other-sector World Bank spending and Chinese agricultural aid. The gains extend beyond yields: aid increases net agricultural income and labor productivity, suggesting that additional output is not simply offset by higher input costs. Consistent with the conceptual framework, greater aid exposure is also associated with increased input purchases, mechanization, crop diversification, and access to information and communication technology, highlighting the financial and knowledge margins through which aid operates.

Who benefits from agricultural aid is as important as its average effect. The largest gains accrue to rural and less-educated households, those facing the strongest barriers to financing modern inputs and accessing agronomic knowledge. This pattern is consistent with aid relaxing binding constraints to technology adoption rather than merely subsidizing farmers who would have adopted improved practices in its absence. The benefits are not concentrated among the highest-yield households: quantile estimates indicate similar proportional gains across the yield distribution. The effects are positive across crop categories, largest for cash crops, and strong for the staple cereals, including rice, millet, sorghum, and maize, the crops central to food security in the region. Gains are also concentrated in districts unaffected by conflict; the absence of a precisely estimated effect in conflict-affected districts reflects weak identifying variation rather than evidence that aid is ineffective in such settings.[1]

The sharpest contrast is between money pledged and money delivered. The yield response is driven by disbursements, the resources that actually reach districts, rather than commitments, the amounts approved on paper. What raises productivity is implementation, not announcement, and this distinction provides the central policy lesson of the paper.

Related literature. This paper contributes to three strands of research: the macroeconomic literature on foreign aid effectiveness, the agricultural-development literature on smallholder productivity constraints in Sub-Saharan Africa, and the emerging literature linking geocoded aid flows to local economic outcomes.

The first strand asks whether foreign aid promotes economic development. The evidence remains contested. Burnside and Dollar (2000) argue that aid is effective under sound policy environments, while Easterly et al. (2004) show that this result is sensitive to sample expansion. Rajan and Subramanian (2008) find no robust aggregate growth effect and document potential Dutch-disease effects through reduced manufacturing competitiveness (Rajan and Subramanian, 2011). More recent studies that address aid allocation more directly recover positive effects (Dreher and Langlotz, 2020), while Clemens et al. (2012) show that aid composition matters, with productive-sector flows generating faster returns than humanitarian assistance. This literature operates primarily at the country level, providing limited evidence on the productivity mechanisms through which aid operates while combining aid categories with different proximate effects. We contribute by isolating a specific category of aid, World Bank agricultural disbursements, and tracing its effect through the farm productivity channel where outcomes are observed.

The second strand examines why agricultural productivity remains low among smallholders in Sub-Saharan Africa. Dercon and Gollin (2014) document the region's slow agricultural transformation and

[1]The leave-one-out instrument has little identifying power in the conflict subsample (Kleibergen–Paap $F = 0.41$). Accordingly, we report these estimates for completeness but do not draw substantive conclusions from them.

persistent yield gaps, while Christiaensen et al. (2011) and the World Bank (2008) show that agriculture-led growth reduces poverty at least twice as effectively as growth elsewhere. A large body of work identifies limited input adoption as a key constraint: Duflo et al. (2008) find substantial but unrealized returns to fertilizer use in Kenya, reflecting liquidity constraints and behavioral barriers. While these studies identify important mechanisms, evidence on whether agricultural aid generates broad productivity gains among West African smallholders remains limited. We contribute by estimating the effect of agricultural aid exposure on farm yields using household-level panel data and an instrumental-variable design.

The third strand exploits geocoded aid data to study how external resources translate into local outcomes. Using project-level aid information assembled by Dreher et al. (2021) and consolidated in the Geocoded Official Development Assistance Dataset (Bomprezzi et al., 2024), recent studies show that aid allocation is shaped by political and geographic factors (Hodler and Raschky, 2014; Dreher et al., 2019). On the outcome side, Dreher et al. (2021) find positive local growth effects of Chinese aid using a Bartik-type instrument, while Martorano et al. (2020) document gains in child education and mortality outcomes in Sub-Saharan Africa. We extend this literature by applying the geocoded-aid approach to World Bank agricultural projects and examining a direct measure of productivity: household crop yields.

The remainder of the paper is organized as follows. Section 2 describes the institutional context of World Bank agricultural aid in West Africa. Section 3 presents the conceptual framework, data, and identification strategy. Section 4 presents the main results, robustness checks, and heterogeneity analysis. Section 5 concludes.

2 Institutional Background

The World Bank is the largest multilateral source of agricultural finance in West Africa. Through its International Development Association (IDA) lending window, it has committed billions of dollars to the eight WAEMU member states (Benin, Burkina Faso, Côte d'Ivoire, Guinea-Bissau, Mali, Niger, Senegal, and Togo) over the past two decades. Because these projects are designed jointly with national governments and implemented at the district and sub-district level, aid flows can be disaggregated geographically at the level required by our identification strategy.

World Bank agricultural projects combine several components, two of which, input-financing and extension, correspond directly to the mechanisms formalised in the conceptual framework of Section 3. Input-financing activities, including subsidies for seeds, fertilizer, and mechanisation, rural credit lines, and matching grants, relax the liquidity constraint that prevents smallholders from purchasing optimal input bundles even when marginal returns are high (Duflo et al., 2008). Extension and information activities, including agronomic advisory services, demonstration plots, and on-farm trials, narrow the gap between the agricultural frontier and the techniques farmers actually use, with potentially larger gains among farmers with lower baseline agronomic knowledge. Infrastructure investments, including irrigation rehabilitation, rural roads, storage, and electrification of agro-processing, provide an additional channel by reducing input and output transaction costs and buffering households against climate and price shocks. Because projects typically combine these components, the empirical estimate is a reduced-form effect capturing their joint impact.

An important institutional distinction in Bank lending is between *commitments*, the funds approved at project initiation, and *disbursements*, the resources actually transferred as implementation milestones are met. Because disbursements lag commitments by months to years and are spread over the implementation period, only the disbursement series captures resources available to affect agricultural production in a given year. We therefore use disbursements as our primary treatment variable and report commitments as a robustness check.

The eight WAEMU countries share a structural context that makes such financing particularly important. Agriculture accounts for 20 to 45 percent of GDP and employs more than 60 percent of the labor force; smallholder farms of one to three hectares dominate the landscape; and cereal yields remain roughly half those observed in South Asia despite broadly comparable agro-ecological conditions (African Development Bank and Food and Agriculture Organization, 2015; World Bank, 2008). Land is largely held under customary tenure and cannot serve as collateral (Platteau, 1996; Goldstein and Udry, 2008), limiting farmers' ability to finance input adoption through private credit markets and making external liquidity, such as aid-financed input support, one of the few channels through which input use can expand at scale. The countries also span three major agro-ecological zones—coastal Guinea, Sudan-Guinean, and Sahelian—that differ in rainfall patterns, crop systems, and exposure to armed conflict. These differences, together with variation in household location and education, provide a natural setting for examining whether the effects of agricultural aid vary across households and local conditions.

3 Conceptual Framework and Empirical Strategy

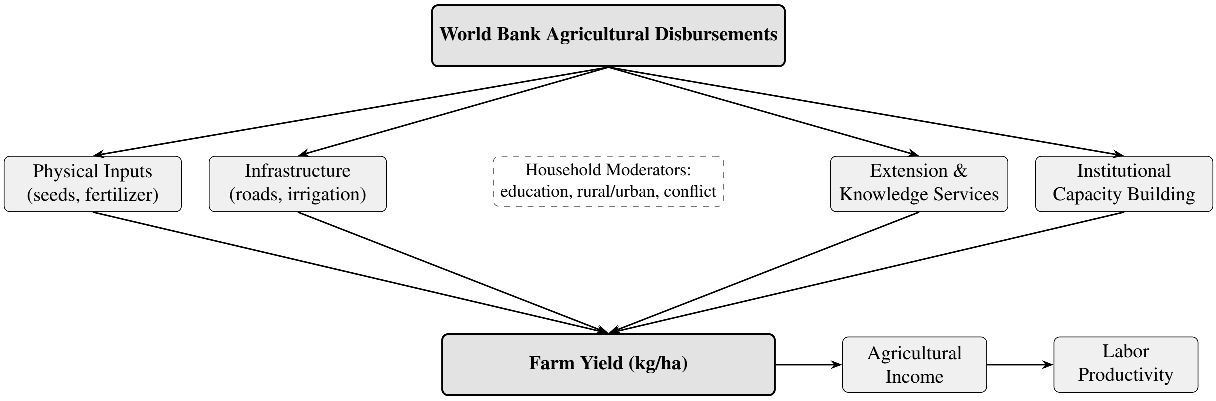
This section develops the framework underlying the empirical analysis. We first present a conceptual model that describes the channels through which agricultural aid can affect farm yields and derives the associated heterogeneity implications. We then describe the data and sample construction, before presenting the identification strategy and instrumental variable approach used to estimate the causal effect of aid.

3.1 Conceptual Framework

Agricultural aid can increase farm productivity by relaxing constraints that limit the adoption of productivity-enhancing technologies. We consider two primary channels. First, aid can relax financial constraints by lowering the cost of modern inputs and improving farmers' access to productive assets. Second, aid can promote technology adoption through extension services, training, and improved access to agronomic information (Foster and Rosenzweig, 1995; Duflo et al., 2008). These mechanisms increase input use and facilitate the adoption of improved production practices, ultimately raising crop yields. Investments in local infrastructure and institutions may reinforce these channels by improving market access and reducing transaction costs. Figure 1 provides a visual overview of the transmission mechanisms.

Next, we present a simple household production model that formalizes the two main channels through which agricultural aid affects productivity. The framework yields two empirical predictions. First, aid should increase agricultural productivity by relaxing constraints that prevent households from reaching their production frontier. Second, the magnitude of these gains should be larger among households facing tighter financial constraints or larger knowledge gaps.

Figure 1: Conceptual Framework: Transmission Channels from Aid to Farm Yield



Notes: This figure illustrates the transmission channels from World Bank agricultural disbursements to farm yield.

Environment. A farming household i in district d at time t produces agricultural output according to:

$$Y_{idt} = A_{idt} f(x_{idt}, e_{idt}), \tag{1}$$

where Y_{idt} denotes crop yield per cultivated hectare, x_{idt} represents purchased inputs (seeds, fertilizer, and hired labour), e_{idt} captures farm management effort, and A_{idt} denotes effective agricultural productivity. The production function is increasing and concave in inputs and effort, with $f_{xe} \geq 0$ allowing for complementarities between input use and management practices. This specification follows the standard agricultural household model (Singh et al., 1986) and reflects evidence that input effectiveness depends on complementary knowledge and management decisions in smallholder agriculture (Duflo et al., 2008).

Normalising the output price to unity, households choose inputs and effort to maximise net agricultural income:

$$\pi_{idt} = Y_{idt} - p_x x_{idt} - C(e_{idt}, \theta_i), \tag{2}$$

where p_x is the input price and θ_i captures household ability, including farming experience, managerial capacity, and human capital. The effort cost function reflects that more able farmers can implement farm management practices at lower cost.[2] Effective productivity A_{idt} combines baseline agronomic knowledge with improvements generated by district-level access to extension services, input markets, and other productivity-enhancing investments.

Financial Constraints and Input Access. Let D_{dt} denote World Bank agricultural disbursements received by district d in year t. Households may face financing constraints that limit their ability to acquire productivity-enhancing inputs. Let κ_i denote household financial capacity, determined by wealth, collateral, and access to credit. Following Banerjee and Duflo (2007), input expenditure is constrained by:

$$p_x x_{idt} \leq \kappa_i + B(D_{dt}), \qquad B'(D_{dt}) > 0, \quad B(0) = 0, \tag{3}$$

[2] The effort cost function captures that households differ in their ability to manage farming activities. We assume $C_e > 0$, $C_{ee} > 0$, $C_\theta < 0$, and $C_{e\theta} < 0$, so that more experienced or skilled farmers face lower effort costs. A standard parametrisation is $C(e, \theta_i) = c(e)/\theta_i$, with $c' > 0$ and $c'' > 0$. These assumptions ensure an interior solution for effort but do not affect the model's predictions regarding the financial and knowledge transfer channels.

where $B(D_{dt})$ captures the improvement in input access generated by aid through mechanisms such as strengthened input supply chains, subsidised procurement schemes, or improved liquidity of implementing agencies. When the constraint binds, households cannot attain their unconstrained input level. Letting $\mu_i \geq 0$ denote the associated shadow value, the first-order conditions are:

$$A_{idt}\, f_x(x^*, e^*) = p_x + \mu_i, \qquad A_{idt}\, f_e(x^*, e^*) = C_e(e^*, \theta_i). \tag{4}$$

When the financial constraint binds, input use is determined by: $x^* = (\kappa_i + B(D_{dt}))/p_x$, and the shadow value of relaxing the constraint is: $\mu_i = A_{idt} f_x(x^*, e^*) - p_x > 0$. Because the marginal product of inputs declines with input use ($f_{xx} < 0$), households with lower financial capacity operate at lower input levels, face larger input wedges, and therefore experience larger yield gains from aid-induced improvements in input access.

Proposition 1 (Financial Channel)**.** *When Equation (3) binds, disbursements raise yields through expanded input access:*

$$\left.\frac{\partial Y_{idt}}{\partial D_{dt}}\right|_{financial} = A_{idt}\, f_x(x^*, e^*) \cdot \frac{B'(D_{dt})}{p_x} > 0. \tag{5}$$

Moreover, this effect is larger among households with lower financial capacity:

$$\frac{\partial^2 Y_{idt}}{\partial D_{dt}\, \partial \kappa_i} = A_{idt}\, f_{xx}(x^*, e^*) \cdot \frac{B'(D_{dt})}{p_x^2} < 0, \tag{6}$$

since $f_{xx} < 0$. Financially constrained households therefore experience larger yield gains from aid-induced improvements in input access.

Proof. Substituting $x^* = (\kappa_i + B(D_{dt}))/p_x$ into the yield function and differentiating with respect to D_{dt} gives Equation (5). Differentiating this expression with respect to κ_i and using $f_{xx} < 0$ gives Equation (6). The Lagrangian formulation and full comparative statics are provided in Appendix A. □

Information, Extension, and Know-How. Aid-financed extension services increase household-level effective productivity A_{idt}. Let A_i^0 denote household i's baseline agronomic knowledge, determined by education, farming experience, and prior access to advisory services. Following the technology diffusion literature (Foster and Rosenzweig, 1995), effective productivity depends on the extent to which aid-funded extension closes the gap between baseline knowledge and the agricultural frontier $A_{\max}$:

$$A_{idt} = A_i^0 + \rho(D_{dt})\big(A_{\max} - A_i^0\big), \qquad \rho'(D_{dt}) > 0, \quad \rho(0) = 0, \tag{7}$$

where $A_{\max}$ denotes frontier agricultural productivity and $\rho(D_{dt}) \in [0,1]$ is the fraction of the knowledge gap closed through aid-financed extension services.

Proposition 2 (Knowledge Transfer Channel)**.** *The yield gain from the knowledge transfer channel is:*

$$\left.\frac{\partial Y_{idt}}{\partial D_{dt}}\right|_{info} = \rho'(D_{dt})\big(A_{\max} - A_i^0\big) f(x^*, e^*) > 0. \tag{8}$$

This effect is decreasing in A_i^0: households further from the agricultural frontier benefit more from aid-funded extension. Since baseline agronomic knowledge is positively correlated with formal education, less educated farmers are expected to experience larger productivity gains from this channel.

Proof. Differentiating $Y_{idt} = A_{idt} f(x^*, e^*)$ with respect to D_{dt} and using Equation (7) yields Equation (8). Differentiating the resulting expression with respect to A_i^0 gives $\partial(\partial Y/\partial D)/\partial A_i^0 = -\rho'(D_{dt}) f(x^*, e^*) < 0$, establishing that the knowledge-transfer effect decreases with baseline agronomic knowledge. The full derivation is provided in Appendix A. □

A common implication emerges from the two mechanisms. Agricultural aid can raise farm productivity by relaxing constraints that prevent smallholder households from reaching the agricultural frontier. Financial support expands access to productive inputs when households face binding liquidity constraints, while extension and information services accelerate the diffusion of improved production practices. These mechanisms operate jointly, with their relative importance depending on household characteristics and project composition. Combining the two channels, the marginal effect of agricultural disbursements on yield can be written as:

$$\frac{\partial Y_{idt}}{\partial D_{dt}} = \underbrace{A_{idt} f_x \cdot \frac{B'}{p_x}}_{\text{financial channel}} + \underbrace{\rho'(D_{dt}) \cdot (A_{\max} - A_i^0) \cdot f}_{\text{knowledge transfer}} > 0, \tag{9}$$

where the first term operates when financial constraints bind and the second term captures productivity gains from closing knowledge gaps. The model therefore predicts three patterns that guide the empirical analysis. First, agricultural aid should increase average yields through the combined effect of these channels. Second, returns should be larger among households facing tighter financial constraints or greater gaps from the agricultural knowledge frontier. Third, these mechanisms provide a framework for interpreting differences in estimated returns across households and districts.

The empirical strategy estimates the average yield effect of agricultural disbursements using an instrumental-variable approach. The heterogeneity analysis then examines whether the estimated effects vary systematically with household and district characteristics that capture differences in financial constraints, access to information, and local economic conditions.

3.2 Data

Our analysis combines nationally representative household surveys, geocoded World Bank project data, and district-level conflict indicators. The household surveys provide farm-level outcomes and characteristics, the project data measure local exposure to World Bank agricultural disbursements, and the conflict data capture local security conditions. These sources are merged at the district (ADM2) and survey-year level. Full variable definitions, harmonization procedures, and sample-construction details are reported in Appendix B.

Household Survey Data. Our household data come from the two waves of the Harmonized Survey on Household Living Conditions (EHCVM), administered by national statistical offices in collaboration with the World Bank across eight West African countries: Benin, Burkina Faso, Côte d'Ivoire, Guinea-Bissau, Mali, Niger, Senegal, and Togo. The first wave covers the 2018/19 agricultural season and the second covers 2021/22. The panel structure allows us to exploit within-household changes in exposure to district-level aid over time, while household fixed effects absorb time-invariant household characteristics. After merging with district-level aid data and excluding observations with missing outcomes or key controls, the regression sample contains 28,726 household-year observations across more than 200 ADM2 districts.

The panel is unbalanced: 91 percent of households observed in 2018 are re-observed in 2021, and household fixed effects are identified from this retained subset. Attrition is not significantly related to baseline district-level aid exposure after controlling for baseline household characteristics and country fixed effects (Table B.2, Appendix C), reducing concerns that selective attrition drives the within-household estimates.

Outcomes and Control Variables. Our primary outcome is agricultural yield, defined as total crop production in kilograms per cultivated hectare and expressed in logarithms. We also examine net agricultural income, labor productivity, and the value of output per hectare (Table 3), and crop-specific yields for cereals, non-cereals, food crops, and cash crops (Table 5), to assess whether the aggregate effect reflects broader productivity gains, holds in value as well as physical terms, and extends across crop types.

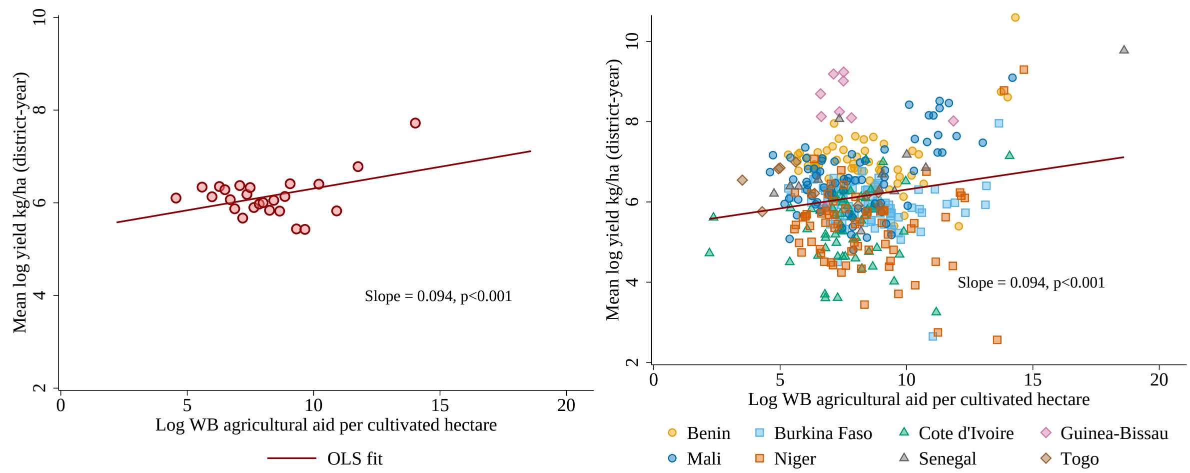
Following standard practice in the agricultural household model literature (Duflo et al., 2008; Christiaensen et al., 2011), our main specifications include time-varying controls for household demographics (household size, head's age and age-squared, gender, and education), farm characteristics (cultivated area, fertilizer use, and mechanization), market access, and local economic conditions. Additional production, technology, and conflict controls are introduced in robustness specifications. Full variable definitions are provided in Appendix B.

World Bank and Chinese Aid Data. District-level aid data come from the Geocoded Official Development Assistance Dataset (GODAD), Version 1.0 (Bomprezzi et al., 2024), which provides geocoded World Bank IBRD/IDA projects and Chinese official development finance. We aggregate project locations to the ADM2 level to construct local aid exposure. Our primary treatment variable is World Bank agricultural disbursements, defined as the financial resources actually transferred to agricultural projects in district d and year t. We distinguish these from commitments, which represent approved project amounts that may be disbursed over subsequent years. Both measures are expressed in constant U.S. dollars using the transformation $\log(\text{Aid}_{dt}+1)$.

GODAD also identifies World Bank projects outside agriculture and Chinese agricultural finance. These variables are included in robustness specifications to verify that the estimated effect is not driven by broader development spending or concurrent agricultural investments from other donors.

Conflict Data. We use the Armed Conflict Location & Event Data (ACLED) project (Raleigh et al., 2010) to construct district-year measures of conflict exposure. Specifically, we calculate the number of violent events targeting civilians, which enters as a control variable, and a binary indicator for any civilian attack, which is used in the heterogeneity analysis.

Descriptive Statistics. Summary statistics for the main variables are reported in Table B.1 (Appendix B). The average log yield is 6.239, corresponding to approximately 512 kg/ha at the geometric mean, with substantial dispersion ($\sigma = 1.785$) reflecting differences in agro-ecological conditions, crop composition, and input access across households and countries. World Bank agricultural disbursements are positive in 65.5% of district-year observations and exhibit substantial right skewness, with a conditional mean of USD 1.57 million compared with a conditional median of USD 0.51 million. This distribution motivates the use of the $\log(\text{Aid}_{dt}+1)$ transformation throughout the analysis.



(a) Binscatter (25 quantile bins)

(b) District-year scatter (country-coded)

Figure 2: Aid and yield in the raw data (district-year)

Notes: This figure shows the unconditional association between agricultural aid and yields at the district-year level, restricted to district-years with positive disbursements. Yield is the average household log(kg/ha) within each district-year cell; the horizontal axis reports $\log(\text{Aid}_{dt}/\text{cultivated area}_{dt})$. Panel (a) presents a 25-quantile binscatter, plotting the mean yield within each bin against mean aid intensity. Panel (b) displays the underlying district-year observations, with marker colour and shape identifying the eight WAEMU countries. The solid red line reports the pooled OLS fit.

Figure 2 provides a descriptive view of the relationship between agricultural aid and farm yields before accounting for potential confounders. The sample is restricted to district-years with positive agricultural disbursements, since aid intensity, defined as the log of disbursements per cultivated hectare, is not defined at zero. Panel (a) plots mean log yield against aid intensity using 25 quantile bins, while Panel (b) displays the underlying district-year observations by country. Within this positive-aid sample, the unconditional relationship is positive: the pooled OLS slope is 0.094 (s.e. 0.028, $p < 0.01$), indicating that district-years with higher aid intensity also exhibit higher yields. This association combines variation across countries, years, and districts and does not account for the non-random allocation of aid. It may therefore reflect differences in agricultural potential or other factors jointly affecting aid placement and productivity rather than a causal effect. Section 3.3 addresses this identification challenge by combining household fixed effects with a leave-one-out instrument to recover the causal effect of agricultural disbursements.

3.3 Identification Strategy

Our baseline estimating equation relates log agricultural yield for household i in district d and survey year t to the log of World Bank agricultural disbursements allocated to district d in year t:

$$Y_{idt} = \alpha_i + \delta_t + \beta\,\text{Aid}_{dt} + \gamma'\mathbf{X}_{idt} + \varepsilon_{idt}, \tag{10}$$

where Y_{idt} is the log of crop yield (kg per cultivated hectare) and Aid_{dt} is the log of World Bank agricultural disbursements in district d at time t.[3] Our preferred specification includes predetermined and time-varying

[3]Because the treatment variable is $\log(\text{Aid}+1)$ rather than $\log(\text{Aid})$, the coefficient β is strictly speaking a semi-elasticity rather than a pure elasticity: it measures the proportional change in yield associated with a unit change in $\log(\text{Aid}+1)$, not in $\log(\text{Aid})$. However, given the large disbursement values in our sample (the conditional mean is USD 1.57 million and the conditional median USD 0.51 million) so $\text{Aid} \gg 1$ for virtually all non-zero observations, so $\log(\text{Aid}+1) \approx \log(\text{Aid})$ and the coefficient is numerically very close to a standard elasticity. We use the term *yield semi-elasticity* throughout for precision.

household- and farm-level controls (household size, household head age and its square, gender, secondary education, log cultivated area, chemical fertilizer use, motorized tillage, distance to the nearest city, religion, urban/rural locality, and district Gini coefficient). Time-invariant household characteristics, such as religion and urban/rural status, are absorbed by α_i and are not separately identified in this specification.

The term α_i denotes household fixed effects, absorbing all time-invariant household heterogeneity, including unobserved managerial ability, soil quality, and geographic endowments; δ_t denotes survey year fixed effects, absorbing shocks common to all households in a given year. With only two survey rounds, β is identified from changes in household yield associated with changes in district-level disbursements between 2018 and 2021, conditional on changes in the included time-varying controls. Standard errors are clustered at the second administrative division (ADM2) district level throughout. The disbursement measure and its timing relative to the agricultural production period are described in Section 3.2 and Appendix B.[4]

Sources of Endogeneity. Two features of the setting could prevent OLS from recovering β. First, aid may be targeted toward districts experiencing weak agricultural conditions, generating correlation between Aid_{dt} and unobserved determinants of yield. Although the direction of this selection bias is not generally signed, targeting toward persistent productivity shortfalls would bias OLS downward. Second, project-level disbursements allocated to districts by geographic coordinates are measured with error; under classical measurement-error assumptions (Bound et al., 2001), this attenuates $\hat{\beta}$ toward zero. These concerns motivate the instrumental-variable strategy.

Instrument Construction and First Stage. We construct a leave-one-out (LOO) instrument, following the finite-sample motivation for excluding own-observation information from instrument construction (Angrist et al., 1999).[5] For each district d in country c and year t, the instrument is the mean of log disbursements across all other districts in the same country-year:

$$Z_{dt} = \frac{1}{|\mathscr{D}_c| - 1} \sum_{d' \in \mathscr{D}_c, d' \neq d} \text{Aid}_{d't}, \tag{11}$$

where $\mathscr{D}_c$ denotes the set of all districts in country c. The instrument is constructed from disbursements to other districts within the same country-year, and is motivated by within-country comovement in World Bank agricultural disbursements: if disbursement timing is correlated across districts within a country-year, a district's own disbursements should be predictable from disbursements to other districts in the same country-year without incorporating them by construction. We formalize the first stage as

$$\text{Aid}_{dt} = \pi Z_{dt} + \alpha_i + \delta_t + \gamma' \mathbf{X}_{idt} + u_{idt}, \tag{12}$$

[4] Disbursements, rather than commitments, are used throughout because only realized transfers can plausibly affect production in a given year; the dating convention that assigns disbursements to the payment year is described in Appendix B.

[5] We define Aid_{dt} as log district disbursements rather than log disbursements per cultivated hectare. Normalizing aid by cultivated area would introduce the same farm-size measure into both the treatment and the outcome, which is already expressed per hectare, while cultivated area also enters separately as a control in Equation (10). More importantly, it would alter the variation captured by the leave-one-out instrument in Equation (11): Z_{dt} averages disbursements across other districts within a country-year, whereas an intensity measure would average aid-to-area ratios across districts with substantially different cultivated areas. This would make the instrument reflect differences in land area as well as the common aid allocation that it is intended to capture. The raw-data preview in Figure 2 uses aid intensity for visualization only.

Instrument relevance requires $\pi \neq 0$; we report the Kleibergen–Paap weak-instrument statistic (Kleibergen and Paap, 2006) in Section 4. Excluding district d from the construction of Z_{dt} removes any mechanical, arithmetic correlation between Z_{dt} and measurement error in district d's own aid measure; it does not address correlation arising from a shared assignment process, for example if geocoding or reporting errors are themselves correlated within a country-year. Within-country dispersion of the LOO instrument depends mechanically on the number of districts N_c.

Identification and Exclusion Restriction. The exclusion restriction requires that Z_{dt} affect household yields in district d only through Aid_{dt}. Because Z_{dt} is constructed from disbursements to other districts within the same country-year, identification therefore depends primarily on the exogeneity of country-year variation in World Bank agricultural disbursements. A variance decomposition confirms this feature of the instrument: 99.94 percent of the variation in Z_{dt} lies between country-year cells, with only 0.06 percent within cells.[6] The variation in Z_{dt} is therefore overwhelmingly concentrated at the country-year level. Causal identification therefore requires that this country-year variation in World Bank agricultural disbursements be unrelated to unobserved changes in agricultural productivity across districts, conditional on year fixed effects. Year fixed effects absorb shocks common across countries in a given year but do not absorb country-specific, time-varying shocks. If World Bank disbursements to a country rise in response to an agricultural crisis or another country-specific shock that independently affects yields in district d, the exclusion restriction is violated. This is the central identifying assumption of the design, and it is not established by the instrument's construction alone.

Given this concentration of variation, country$\times$year fixed effects would absorb nearly all of the variation in Z_{dt} by construction. Because these fixed effects would leave only the small cross-district variation induced by the leave-one-out construction, we rely instead on household and year fixed effects, which preserve the country-year variation underlying the first-stage relationship.

Taken together, the design uses cross-district comovement in World Bank disbursement timing within countries to instrument district-level aid. The IV coefficient therefore identifies the effect of the component of district-level disbursements predicted by contemporaneous disbursement patterns across other districts in the same country. The empirical results assess instrument relevance and the sensitivity of the estimates to the principal threats to the exclusion restriction.

4 Results

This section presents the empirical findings in four steps. We first report the main IV estimates and assess their economic magnitude. We then assess robustness, including alternative treatment measures, alternative outcome variables, and sensitivity to functional form, geography, and inference. We next turn to heterogeneity, asking which households benefit most. Finally, we validate the mechanisms through which aid raises yields.

[6] This concentration follows mechanically from the leave-one-out construction: for two districts d and d' in the same country-year, $Z_{dt} - Z_{d't} = (\text{Aid}_{d't} - \text{Aid}_{dt})/(N_c - 1)$, so within-cell dispersion in Z_{dt} shrinks as the number of districts N_c in a country grows. Supporting diagnostics, including the relationship between within-cell dispersion and district count across country-year cells, are reported in Appendix B.7.

4.1 Main Estimates

Table 1 reports the main estimates of the effect of agricultural aid on household crop yields. Columns (1) and (2) report OLS estimates, while columns (3) and (4) instrument district-level agricultural disbursements using the leave-one-out (LOO) instrument described in Section 3.3. Columns (1) and (3) include household and year fixed effects and constitute the preferred within-household specifications; columns (2) and (4) replace household fixed effects with country fixed effects. All specifications include the full set of household, farm, and district controls, with standard errors clustered at the ADM2 district level.

Table 1: Main Results: OLS and IV

	(1) OLS	(2) OLS	(3) IV	(4) IV
Aid_{dt}	0.021^{**}	0.010^{*}	0.127^{***}	0.218^{**}
	(0.009)	(0.005)	(0.042)	(0.089)
Controls	✓	✓	✓	✓
Household FE	✓		✓	
Country FE		✓		✓
Year FE	✓	✓	✓	✓
Observations	28,726	36,673	28,726	36,673
Within R^2	0.221	0.232		
KP F-stat			18.49	7.09

Notes: This table presents the effect of World Bank agricultural disbursements on log household yield (kg/ha). The treatment in all columns is $\log(\text{Aid}_{dt})$. Columns (1)–(2) are OLS; columns (3)–(4) are IV using the leave-one-out (LOO) district instrument. Columns (1) and (3) include household and year fixed effects; columns (2) and (4) include country and year fixed effects. Column (3) is the preferred IV specification. All columns include the full set of household and farm controls (cultivated area, chemical fertilizer use, motorized tillage, household size, head's age and its square, head's gender, secondary education, distance to nearest city, district Gini coefficient, and an urban/rural locality indicator), whose coefficients are suppressed for readability. Standard errors clustered at the ADM2 district level in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The preferred IV estimate in column (3) is positive and precisely estimated. The coefficient of 0.127 ($p < 0.01$) implies that a 10 percent increase in agricultural aid disbursements raises household crop yields by approximately 1.3 percent. The corresponding OLS estimate is 0.021 ($p < 0.05$), positive and significant in its own right but six times smaller than the IV estimate. This gap is consistent with the non-random allocation of agricultural aid toward districts with poorer agricultural conditions, together with measurement error in district-level disbursements, both of which attenuate OLS toward zero. The first-stage Kleibergen–Paap *rk* Wald F-statistic is 18.49, well above the conventional threshold of 10 (Stock and Yogo, 2005), indicating that the preferred specification does not raise a conventional weak-instrument concern.

The magnitude of the estimate is economically meaningful. At the geometric mean yield of approximately 512 kg per hectare, a doubling of agricultural disbursements is associated with an additional 47 kg per hectare.[7] For comparison, this increase represents approximately one-fifth of the yield difference

[7]The geometric mean, $\exp(\overline{\log y}) \approx 512$ kg/ha, is computed from the mean of log yield reported in Table B.1 ($\overline{\log y} = 6.239$). We use the geometric rather than the arithmetic mean of yield in levels (4,894 kg/ha) because the level distribution is heavily right-skewed (skewness ≈ 120), with a small number of extreme outliers. The geometric mean is close to the median (521 kg/ha) and provides a more representative baseline for a typical household. The reported 47 kg/ha is the exact effect implied by the

associated with chemical fertilizer use in the sample ($\hat{\beta}_{\text{fert}} = 0.392$, coefficient suppressed from Table 1 for readability).

The results are similar in sign when household fixed effects are replaced by country fixed effects, although the magnitude of the IV estimate is larger. Column (4) yields a coefficient of 0.218 (s.e. 0.089, $p < 0.05$), while the corresponding Kleibergen–Paap *rk* Wald F-statistic is 7.09. Because this specification relies on a smaller source of identifying variation and provides weaker first-stage evidence, we treat column (3) as the preferred IV specification.

The contrast between the OLS and IV estimates is also consistent with the broader aid-effectiveness literature. Cross-country studies have generally found limited evidence of an effect of aggregate aid on growth (Rajan and Subramanian, 2008), while studies distinguishing productive and early-impact aid find more positive effects (Clemens et al., 2012; Dreher and Langlotz, 2020). Our findings complement this literature by focusing on a specific productive sector and exploiting subnational variation in actual aid disbursements. The fact that the OLS estimate is already positive, while the IV estimate is substantially larger, is consistent with the importance of accounting for the endogenous allocation of aid to local conditions (Hodler and Raschky, 2014; Dreher et al., 2019).

4.2 Robustness

This section examines the robustness of the baseline result along three dimensions: alternative measures of aid exposure, alternative measures of farm productivity, and sensitivity to functional form, geographic controls, and randomization-based inference.

Alternative Treatment Measures. Table 2 examines whether the baseline result depends on the measurement or specification of aid exposure. Column (1) replaces continuous disbursements with a binary indicator equal to one if at least one World Bank agricultural project was active in district d and year t. Column (2) uses the number of active agricultural projects as the treatment. Column (3) replaces disbursements with log commitments, which capture funds pledged rather than resources actually delivered. Columns (4) and (5) retain the baseline disbursement measure while controlling, respectively, for disbursements from other World Bank sectors and for Chinese agricultural aid commitments.

The extensive-margin specification in column (1) yields a positive and statistically significant coefficient, indicating that the baseline result is also present when aid exposure is defined by whether a district has at least one active agricultural project. Column (2) likewise yields a positive coefficient on the number of active projects, indicating that greater project exposure is associated with higher yields. These specifications therefore suggest that the baseline result is not an artifact of using the continuous volume of disbursements as the treatment measure.

Column (3) provides a complementary test using aid commitments rather than disbursements. The coefficient on log commitments is substantially smaller than the corresponding coefficient on log disbursements in column (3) of Table 1. This pattern suggests that actual resource delivery is more closely associated with yield gains than the amount of funding committed. We interpret this result as evidence that the timing and delivery of resources matter for the estimated relationship, rather than as a direct test of the mechanism through which aid affects productivity.

estimated semi-elasticity, $512 \times (2^{\hat{\beta}} - 1) \approx 47$ kg/ha; the corresponding linear approximation, $512 \times \hat{\beta} \times \log(2)$, is approximately 45 kg/ha. The fertilizer comparison is based on the ratio of the estimated semi-elasticities, $(\hat{\beta} \log 2)/\hat{\beta}_{\text{fert}}$, and is therefore independent of the choice of baseline.

Table 2: Alternative Treatment Measures

	(1) Binary	(2) Count	(3) Commit.	(4) +Sectors	(5) +China
Any WB agricultural aid (0/1)	2.303***				
	(0.767)				
Number of WB projects		0.495***			
		(0.190)			
WB agricultural commitments (log)			0.044***		
			(0.017)		
Aid_{dt}				0.181***	0.127***
				(0.070)	(0.042)
Controls	✓	✓	✓	✓	✓
Household FE	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓
Observations	28,726	28,726	28,726	28,726	28,726
KP F-stat	12.42	27.77	59.23	13.61	18.35

Notes: This table presents IV estimates of the yield effect across alternative measures of aid exposure. Columns (1)–(2) use alternative measures of agricultural aid exposure: an indicator for an active project and the number of active projects. Column (3) replaces agricultural disbursements with commitments. Columns (4)–(5) retain baseline agricultural disbursements and additionally control for other World Bank sectoral disbursements and Chinese agricultural aid commitments, respectively. Standard errors are clustered at ADM2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Columns (4) and (5) address the possibility that the estimated effect of agricultural aid captures other forms of external assistance. Controlling for other World Bank sectoral disbursements yields $\hat{\beta} = 0.181$ (s.e. 0.070, $p < 0.01$), while controlling for Chinese agricultural aid leaves the coefficient essentially unchanged at $\hat{\beta} = 0.127$ (s.e. 0.042, $p < 0.01$). The stability of the estimate when Chinese agricultural aid is included is particularly reassuring given the possibility of correlated aid allocation across donors. The somewhat larger estimate in column (4) is also consistent with the negative correlation between agricultural and non-agricultural World Bank disbursements in the sample. Overall, these exercises indicate that the baseline relationship is not sensitive to alternative measures of aid exposure or to the inclusion of other major sources of external assistance.

Alternative Outcomes. Table 3 examines whether the yield response in Table 1 extends to broader measures of farm productivity and economic performance. The three columns use, respectively, log net agricultural income per hectare, log labor productivity (measured as the value of output per day of family labor), and log output value per hectare.

Column (1) shows a positive and statistically significant effect on net agricultural income per hectare: the increase in physical yields documented in the baseline specification is accompanied by higher net agricultural returns. Column (2) similarly indicates a positive effect on labor productivity, suggesting that the increase in agricultural output is not simply proportional to the amount of family labor employed. Column (3) shows that the result also holds when agricultural production is valued at local market prices rather than measured in physical units: $\hat{\beta} = 0.280$ (s.e. 0.075, $p < 0.01$), providing additional evidence that the baseline yield result is not driven mechanically by changes in the physical composition of production.

These findings extend the baseline result beyond physical crop yields to income, labor productivity, and the market value of agricultural production, and are broadly consistent with the geocoded-aid literature documenting positive local effects of external assistance on economic activity (Dreher et al., 2021) and child education and mortality outcomes (Martorano et al., 2020). Our results provide complementary

Table 3: Alternative Outcomes

	(1) Net income	(2) Labor productivity	(3) Output value/ha
Aid_{dt}	0.235***	0.234**	0.280***
	(0.062)	(0.099)	(0.075)
Controls	✓	✓	✓
Household FE	✓	✓	✓
Year FE	✓	✓	✓
Observations	25,106	21,364	28,720
KP F-stat	19.90	8.34	18.48

Notes: This table presents IV estimates of the effect of agricultural disbursements on alternative outcome measures. Column (3), output value per hectare, is total crop production valued at local market prices and divided by cultivated area, as opposed to physical yield in kilograms per hectare in Table 1. Standard errors are clustered at ADM2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

evidence at the farm level, linking World Bank agricultural assistance to measurable changes in agricultural productivity.

One qualification concerns column (2): the Kleibergen–Paap F-statistic for the labor-productivity specification is slightly below the conventional threshold of 10 in the full sample, so the corresponding 2SLS estimate should be interpreted with greater caution than the other columns. Nevertheless, its positive sign and consistency with the broader pattern provide suggestive evidence that the productivity response extends to labor efficiency.

Sensitivity to Functional Form, Geography, and Inference. We next assess the sensitivity of the preferred IV specification in column (3) of Table 1 to alternative functional forms, finer geographic controls, and randomization-based inference. The corresponding results are reported in Appendix C; the main findings are summarized below.

First, we introduce locality (village) fixed effects and locality-by-year fixed effects to absorb increasingly fine geographic sources of heterogeneity (Table C.4). The coefficient remains positive and statistically significant under both specifications: $\hat{\beta} = 0.127$ ($p < 0.01$) with locality fixed effects and $\hat{\beta} = 0.188$ ($p < 0.01$) with locality-by-year fixed effects. The stability of the estimate when locality-by-year effects are included is particularly informative because these effects absorb shocks common to all households within a locality in a given survey year, including differential exposure to the COVID-19 period in the 2021 wave.

Second, the estimate is similar under alternative functional forms and sample restrictions. Replacing the log treatment with the inverse hyperbolic sine transformation yields $\hat{\beta} = 0.119$ ($p < 0.01$), while trimming observations with extreme yields produces $\hat{\beta} = 0.096$ ($p < 0.01$; Table C.5). The persistence of the positive coefficient across these specifications indicates that the baseline estimate is not driven by the logarithmic functional form or by a small number of extreme yield observations.

Third, we conduct two randomization-inference placebo exercises (Table C.6). In the first, district-level aid is randomly reassigned within country-year cells, preserving the country-year distribution of aid while breaking its alignment with the observed district allocation; the resulting permutation p-value is 0.018. In the second, the instrument is randomly reassigned across district-year observations, producing a permutation p-value below 0.002. In both exercises, the placebo distributions are centered near zero, whereas the baseline estimate lies in their tails. These results provide additional evidence against the possibility that the estimated relationship is driven by the particular alignment between the observed aid allocation and district outcomes.

4.3 Heterogeneity in Treatment Effects

The main estimate establishes that agricultural aid raises average yields. This section investigates which households benefit most and whether those gains are concentrated in particular crops, using the conceptual framework developed in Section 3.1 as an organising lens. The mechanisms are not mutually exclusive, and any given disbursement may activate several simultaneously; the heterogeneity analysis therefore provides an empirical assessment of which mechanisms appear most operative in this sample and setting.

We investigate these questions with two complementary approaches to heterogeneity and a third specification check. The first examines heterogeneity across household and district characteristics by comparing average effects across mutually exclusive subsamples; the second examines whether the aggregate yield response is concentrated in particular crop categories; the third uses an IV quantile specification as a diagnostic for the conditional yield distribution.

Subsample IV Estimates. The financial-constraint mechanism predicts larger returns among households with more limited access to external finance and complementary market infrastructure; we use rural residence as a broad proxy for this dimension rather than a direct measure of collateral access under customary tenure. The knowledge-transfer mechanism predicts larger returns among households farther from the agronomic knowledge frontier, proxied by limited formal education, though this gradient depends on how uniformly project extension services are designed to reach all smallholders. We also explore whether returns vary across conflict-affected districts, where private substitutes for inputs and infrastructure are most scarce; whether such variation is identifiable depends on instrument power within that subsample.

Table 4 presents the subsample IV estimates. Columns (1)–(2) compare rural and urban households (financial constraint dimension). Columns (3)–(4) compare households without and with secondary education (knowledge transfer dimension). Columns (5)–(6) compare conflict-affected and unaffected district-years. To control the false discovery rate across these six tests, we apply the Benjamini-Hochberg correction (Benjamini and Hochberg, 1995) with target rate $q = 0.10$; the final row of Table 4 reports the resulting BH-adjusted p-values, where a value below 0.10 indicates significance after controlling the false discovery rate.

Rural households (column 1) exhibit a larger point estimate than urban households (column 2), consistent with the financial-constraint mechanism. The low-education subsample (column 3) similarly delivers a larger point estimate and is more precisely estimated than the secondary-education subsample (column 4), consistent with the knowledge-transfer mechanism. The conflict comparison reveals a different pattern: the estimated effect is positive but statistically insignificant in conflict-affected districts (column 5), whereas it is positive and statistically significant in non-conflict districts (column 6), with $\hat{\beta} = 0.135$ (KP F-stat $= 11.87$, $p < 0.05$). Thus, the evidence of a positive yield response to agricultural aid is concentrated in non-conflict districts. The conflict estimate is highly imprecise, however, with a standard error of 0.422 and a first-stage F-statistic of only 0.41, so the absence of statistical significance in conflict-affected districts should not be interpreted as evidence of a smaller treatment effect.[8]

[8] After adjustment for multiple testing, the evidence that the rural, urban, low-education, and no-conflict coefficients

Table 4: Heterogeneity: Subsample IV Estimates

	(1) Rural	(2) Urban	(3) Low educ.	(4) Sec. educ.	(5) Conflict	(6) No conflict
Aid_{dt}	0.134***	0.094**	0.130***	0.019	0.277	0.135**
	(0.047)	(0.039)	(0.042)	(0.068)	(0.422)	(0.061)
Controls	✓	✓	✓	✓	✓	✓
Household FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
BH q-value	0.015	0.031	0.013	0.777	0.618	0.044
Observations	24,656	3,996	25,002	1,528	4,340	16,050
KP F-stat	16.46	20.28	19.29	8.20	0.41	11.87

Notes: This table presents subsample IV estimates of the yield effect. Each column reports IV-LOO with household and year fixed effects on the indicated subsample. Subsamples are ordered by mechanism: financial constraints (cols. 1–2), knowledge transfer (cols. 3–4), conflict resilience (cols. 5–6). "Low educ." denotes household heads without secondary education (no formal schooling or primary education only); "Sec. educ." denotes household heads with secondary education or above. The "BH-adjusted p-value" row reports Benjamini-Hochberg adjusted p-values across all six subsample tests (Benjamini and Hochberg, 1995); a value below 0.10 indicates significance after controlling the false discovery rate at $q = 0.10$. Subsamples where KP $F < 10$ should be interpreted with caution. Standard errors clustered at ADM2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The larger point estimates for rural and less-educated households are broadly consistent with the agricultural-development literature, which emphasizes the importance of agricultural gains for rural and poorer populations (Christiaensen et al., 2011; World Bank, 2008). This pattern is consistent with the broader view that agricultural interventions can generate particularly large benefits among populations with greater reliance on agriculture and fewer economic opportunities outside the sector.

Crop-Type Heterogeneity. A complementary cut asks whether the aggregate yield gain is concentrated in particular crops. Table 5 re-estimates the preferred specification separately by crop category: cereals (rice, millet, sorghum, and maize); non-cereal food crops (groundnut, cowpea, okra, sorrel, and watermelon); all food crops (cereals and non-cereal food crops combined); and cash crops.

Table 5: Heterogeneity: Crop-Type Yield Effects

	(1) Cereals	(2) Non-cereals	(3) Food crops	(4) Cash crops
Aid_{dt}	0.191***	0.145	0.162**	0.393**
	(0.070)	(0.114)	(0.066)	(0.189)
Controls	✓	✓	✓	✓
Household FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Observations	23,416	10,082	24,490	5,280
KP F-stat	18.52	7.83	18.20	7.03

Notes: This table presents IV-LOO estimates of the yield effect by crop category. Details on crop-category definitions are given in Appendix B.2. Cash crops and cereals are mutually exclusive categories in this classification. Standard errors clustered at ADM2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

differ from zero remains ($p_{\text{BH}} = 0.015$, 0.031, 0.013, and 0.044, respectively), whereas the secondary-education and conflict coefficients do not ($p_{\text{BH}} = 0.777$ and 0.618). The conflict result is particularly uninformative on identification grounds: its first-stage Kleibergen-Paap F-statistic is 0.41, far below any threshold associated with reliable instrumentation, so this subsample should not be read as evidence either for or against a conflict-related gradient.

The crop-level results indicate that the positive yield response is not confined to a particular type of crop. Cereals (column 1) show a positive and precisely estimated effect, while the pooled food-crop estimate (column 3) is also positive and statistically significant. This suggests that the aggregate yield response is driven in substantial part by food crops, including staple cereals, rather than by a narrow set of specialized crops. The cash-crop estimate (column 4) is larger in magnitude, pointing to a potentially stronger response among commercially oriented crops, although this pattern is less precisely estimated. The estimate for non-cereal food crops (column 2) is positive but not statistically significant. Overall, the crop-level results provide the clearest evidence of positive yield effects for cereals and food crops more broadly, while also suggesting a potentially larger response for cash crops.

The larger cash-crop coefficient is also consistent with the market-access channel discussed in Section 4.4. Because cash crops in the sample are grown predominantly for sale rather than home consumption, improvements in market access may allow commercially oriented farmers to respond more strongly through greater input use, crop choice, or technology adoption. Consistent with this interpretation, Section 4.4 shows that agricultural aid raises the market-and-institutions index, with commercialization cooperatives emerging as a significant channel in two of four specifications (Appendix D). The larger cash-crop response may therefore reflect the greater scope for market-oriented production to benefit from these improvements. This interpretation remains suggestive, however, since the crop-specific estimates do not directly identify the market-access mechanism and the outcome is physical yield rather than revenue or commercialization.

Distributional Specification Check. As a complementary specification check, Table 6 applies an IV quantile regression to examine the conditional yield distribution after accounting for household and year fixed effects, following Chernozhukov and Hansen (2005) and the location-shift approach to fixed effects in Canay (2011). We partial out household and year fixed effects from all variables prior to estimation (the location-shift transformation of Canay, 2011) and then apply instrumental-variable quantile regression at $q \in \{0.25, 0.50, 0.75, 0.90\}$.

Table 6: Distributional Effects: IV Quantile Regression

	(1) q = 0.25	(2) q = 0.50	(3) q = 0.75	(4) q = 0.90
Aid_{dt}	0.125***	0.125***	0.125***	0.125***
	(0.007)	(0.006)	(0.007)	(0.009)
Intercept	-0.502***	-0.000	0.502***	1.036***
	(0.009)	(0.006)	(0.009)	(0.015)
Controls	✓	✓	✓	✓
Quantile	q = 0.25	q = 0.50	q = 0.75	q = 0.90
Observations	28,726	28,726	28,726	28,726

Notes: This table presents IV quantile regression estimates of the distributional effects of agricultural aid on farm yield, following Chernozhukov and Hansen (2005) with the location-shift approach to fixed effects in Canay (2011). Household and year fixed effects are partialled out prior to estimation (location-shift assumption; Canay, 2011). The instrument is the leave-one-out mean of World Bank agricultural disbursements. All specifications include the full baseline control vector (cultivated area, chemical fertilizer use, motorized tillage, household size, head's age and its square, head's gender, secondary education, distance to nearest city, religion, and district Gini coefficient), whose coefficients are suppressed for readability. Standard errors are bootstrap (500 replications). * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Under the location-shift assumption of Canay (2011), the slope coefficient on Aid_{dt} is invariant across quantiles by construction; this specification therefore is a diagnostic for whether the estimated relationship is well-behaved after partialling out fixed effects. Distributional information here is contained in the quantile-varying intercepts, which trace the shape of the yield distribution across quantiles net of fixed effects. The estimated slope is 0.125 across all four quantile levels, and the quantile-specific intercepts rise monotonically from -0.502 at $q = 0.25$ to $+1.036$ at $q = 0.90$, consistent with an orderly, well-identified conditional distribution rather than an implausible or reversed pattern that would signal a problem with the estimation procedure.

4.4 Mechanism Validation

The heterogeneity results in Section 4.3 indicate that the yield response is particularly pronounced among rural and less-educated households. We next examine whether the intermediate outcomes implied by the conceptual framework in Section 3.1 move in ways consistent with the proposed mechanisms. Rather than treating any single intermediate outcome as definitive evidence of a mechanism, we organize the outcomes into three mechanism-specific families and summarize each family using a standardized effects index following Anderson (2008). We then estimate the preferred IV specification, including household and year fixed effects and the full baseline control vector, with each index as the outcome.

Table 7: Mechanism Validation: Standardized Effects Indices

	(1) Financial	(2) Technology & Info.	(3) Market & Institutions
Aid_{dt}	0.034***	0.028***	0.031**
	(0.012)	(0.008)	(0.015)
Controls	✓	✓	✓
Household FE	✓	✓	✓
Year FE	✓	✓	✓
Observations	28,964	26,696	26,484
KP F-stat	18.11	18.48	18.36

Notes: This table presents IV-LOO estimates of the effect of World Bank agricultural disbursements on three standardized effects indices (Anderson, 2008), one for each mechanism. The financial index aggregates the three financial-constraint outcomes in Table D.7; the technology and information index aggregates the six outcomes in Table D.8; the market and institutions index aggregates the five outcomes in Table D.9. Each outcome is standardized by subtracting its sample mean and dividing by its sample standard deviation. The index is then constructed as a weighted average using Anderson's generalized least squares weights, which down-weight outcomes that are highly correlated with the rest of their family. All specifications include household and year fixed effects and the full baseline control vector. Standard errors clustered at ADM2 in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7 reports the resulting estimates. Agricultural disbursements have positive and statistically significant effects on all three indices: the financial index increases by 0.034 standard deviations, the technology and information index by 0.028 standard deviations, and the market and institutions index by 0.031 standard deviations. The results therefore provide evidence that agricultural aid affects intermediate outcomes spanning all three channels proposed in the framework, consistent with the heterogeneous yield effects documented in Section 4.3.

The full estimation results for the fourteen individual outcomes underlying these indices are reported in Appendix D (Tables D.7–D.9). Six outcomes are positive and statistically significant under both OLS and IV: any input purchased, agricultural financing through own funds or credit, crop diversification, ICT access, mechanized farming, and an active land rental market. These results provide more detailed evidence on the specific margins through which the three mechanism indices operate.

The financial outcomes are particularly consistent with the role of liquidity and financing constraints in agricultural production. Duflo et al. (2008) show that fertilizer adoption can remain low despite high returns because of liquidity constraints and present bias, while Karlan et al. (2014) show that relaxing credit and risk constraints can increase agricultural investment. Our finding that aid increases input purchases, input acquisition financed through own resources or credit, and mechanization is consistent with this evidence. More broadly, the results suggest that agricultural aid can relax financial and complementary production constraints that limit farmers' ability to adopt productive inputs and technologies.

5 Conclusion

This paper estimates the effect of World Bank agricultural disbursements on farm yields using a harmonized panel of approximately 29,000 households from eight West African countries observed in 2018 and 2021. The identification strategy exploits a leave-one-out instrument that isolates within-country, between-district variation in agricultural aid allocations. The preferred IV estimate is 0.127, implying that a 10 percent increase in district disbursements is associated with approximately a 1.3 percent increase in household yields. The corresponding OLS estimate is substantially smaller, consistent with negative selection of aid toward less productive districts and attenuation from measurement error in district-level aid measures.

The results point to a coherent set of patterns. The positive yield response is robust to alternative measures of aid exposure, outcomes, functional forms, geographic controls, and randomization-based inference. The response is also evident when agricultural productivity is measured through net agricultural income, labor productivity, and the value of output per hectare. Across household groups, the largest point estimates are found among rural and less-educated households, while the positive and statistically significant effect is concentrated in non-conflict districts. The crop-level evidence is strongest for cereals and food crops more broadly, suggesting that the aggregate yield response is closely connected to staple agricultural production. The results further show that aid increases intermediate outcomes associated with all three channels in the conceptual framework: financial capacity, technology and information, and market and institutional access. Taken together, these findings suggest that agricultural aid raises productivity through a combination of improved access to productive inputs and finance, greater adoption of technologies and information, and stronger market and institutional connections.

Several qualifications are important. First, the instrument identifies the effect of aid allocated to a district and does not separately identify potential spillovers to neighboring districts. If agricultural projects generate positive spillovers across district boundaries, the district-level estimate may understate the broader effect of aid. Second, the data cover only two survey waves and eight countries, limiting what can be learned about longer-run dynamics and restricting the extent to which the estimates can be generalized beyond the West African setting studied here. Third, the conflict subsample provides little identifying variation, so the absence of a precisely estimated effect in conflict-affected districts should not be interpreted as evidence that aid is ineffective in such environments.

The broader implication is that the effectiveness of agricultural aid depends not only on how much funding is committed, but on whether resources reach farmers and relax the constraints that limit their ability to translate those resources into production. The distinction between commitments and disbursements is particularly important: resources that remain pledged but undelivered cannot generate the same production response as funds that reach agricultural projects and households. The concentration of the clearest effects among rural households and in staple food production further suggests that agricultural assistance can have particularly important productivity implications where access to private finance, productive inputs, and markets is limited. For donors and policymakers, the central lesson is therefore less about expanding aid mechanically than about improving its delivery and design: ensuring timely disbursement, strengthening the input and information systems through which farmers respond to assistance, and directing support toward the production systems and households where these constraints are most binding.

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Online Appendix

Does Agricultural Aid Raise Farm Productivity? Evidence from West Africa

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A Derivation of Optimality Conditions

This appendix derives the full optimality conditions for the household problem presented in Section 3.1. It constructs the Lagrangian, derives the Karush-Kuhn-Tucker (KKT) conditions, analyses the constrained and unconstrained cases, establishes the comparative statics underlying Equation (9), and verifies the second-order conditions.

A.1 The Household Problem

Household i in district d at time t chooses purchased inputs $x \geq 0$ and effort $e \geq 0$ to solve:

$$\max_{x\geq 0,\, e\geq 0} \quad \pi_{idt} \;=\; A_{idt}\, f(x,e) \;-\; p_x x \;-\; \frac{c(e)}{\theta_i} \tag{A.1}$$

subject to the financial constraint:

$$x \;\leq\; \frac{\kappa_i + B(D_{dt})}{p_x} \;\equiv\; \bar{x}_i \tag{A.2}$$

where $A_{idt} = A_i^0 + \rho(D_{dt})(A_{\max} - A_i^0)$ as defined in Equation (7), and $\bar{x}_i$ is the maximum input quantity affordable given household financial capacity κ_i and aid-induced credit relief $B(D_{dt})$. The output price is normalised to unity throughout.

The following regularity conditions are maintained throughout:

i) $f : \mathbb{R}^2_+ \to \mathbb{R}_+$ is twice continuously differentiable, strictly increasing and strictly concave: $f_x > 0$, $f_e > 0$, $f_{xx} < 0$, $f_{ee} < 0$, and $f_{xx} f_{ee} - f_{xe}^2 > 0$ (strict concavity).

ii) $c : \mathbb{R}_+ \to \mathbb{R}_+$ is twice continuously differentiable with $c(0) = 0$, $c' > 0$, and $c'' > 0$.

iii) $A_{idt} > 0$, $\theta_i > 0$, $p_x > 0$, $\kappa_i \geq 0$, and $B(D_{dt}) \geq 0$.

iv) An interior solution for effort exists: the Inada conditions $\lim_{e\to 0} f_e = +\infty$ and $\lim_{e\to\infty} f_e = 0$ ensure $e^* > 0$ at any finite A_{idt}.

Associating the multiplier $\mu_i \geq 0$ with Equation (A.2), the Lagrangian is:

$$\mathscr{L}(x,e,\mu_i) \;=\; A_{idt}\, f(x,e) \;-\; p_x x \;-\; \frac{c(e)}{\theta_i} \;+\; \mu_i[\bar{x}_i - x] \tag{A.3}$$

A.2 First-Order and KKT Conditions

The KKT conditions for Equations (A.1)–(A.2) are necessary and sufficient for a global maximum, since $\mathscr{L}$ is strictly concave in (x,e) for any fixed $\mu_i \geq 0$, as verified in Section A.6. Differentiating $\mathscr{L}$ with respect to x yields the stationarity condition for inputs:

$$\frac{\partial \mathscr{L}}{\partial x} = A_{idt} f_x(x^*, e^*) - p_x - \mu_i = 0 \tag{A.4}$$

Differentiating with respect to effort e gives the analogous condition for the effort margin:

$$\frac{\partial \mathscr{L}}{\partial e} = A_{idt} f_e(x^*, e^*) - \frac{c'(e^*)}{\theta_i} = 0 \tag{A.5}$$

Primal feasibility requires $x^* \leq \bar{x}_i$ and dual feasibility requires $\mu_i \geq 0$. The complementary slackness condition then imposes that at least one of these must hold with equality:

$$\mu_i (\bar{x}_i - x^*) = 0 \tag{A.6}$$

Rearranging Equation (A.4), the Lagrange multiplier satisfies:

$$\mu_i = A_{idt} f_x(x^*, e^*) - p_x \tag{A.7}$$

This quantity measures the wedge between the marginal value product of inputs and their market price. When $\mu_i > 0$, the financial constraint is binding and the household cannot attain the first-best input level.

A.3 Case 1: Unconstrained Optimum ($\mu_i = 0$)

When $\mu_i = 0$, Equation (A.6) is satisfied with $x^* < \bar{x}_i$ (the constraint does not bind). Equations (A.4) and (A.5) reduce to:

$$A_{idt} f_x(x^{FB}, e^{FB}) = p_x, \tag{A.8}$$

$$A_{idt} f_e(x^{FB}, e^{FB}) = \frac{c'(e^{FB})}{\theta_i}, \tag{A.9}$$

where Equation (A.8) equates the marginal value product of inputs to the input price, and Equation (A.9) equates the marginal value product of effort to its marginal disutility adjusted for ability. The first-best solution (x^{FB}, e^{FB}) is determined jointly by Equations (A.8) and (A.9). By the implicit function theorem and the strict concavity of f, a unique interior solution exists.

In the unconstrained case, a marginal increase in D_{dt} has no direct effect on (x^{FB}, e^{FB}) through the financial channel because the constraint is slack. The yield response operates exclusively through the knowledge transfer channel (via A_{idt}), derived in Section A.5.

A.4 Case 2: Constrained Optimum ($\mu_i > 0$)

When $\mu_i > 0$, Equation (A.6) requires:

$$x^* = \bar{x}_i = \frac{\kappa_i + B(D_{dt})}{p_x} \tag{A.10}$$

The optimal input level is fully determined by the financial constraint. Effort is still chosen at an interior optimum satisfying Equation (A.5), which can be written as:

$$A_{idt}\, f_e\left(\frac{\kappa_i + B(D_{dt})}{p_x},\, e^*\right) \;=\; \frac{c'(e^*)}{\theta_i} \tag{A.11}$$

Equation (A.11) implicitly defines $e^* = e^*(A_{idt}, \theta_i, x^*)$. Substituting Equation (A.10), optimal yield is:

$$Y^*_{idt} \;=\; A_{idt}\, f\left(\frac{\kappa_i + B(D_{dt})}{p_x},\, e^*\right) \tag{A.12}$$

Shadow Price and Its Determinants. From Equations (A.7) and (A.10):

$$\mu_i \;=\; A_{idt}\, f_x\left(\frac{\kappa_i + B(D_{dt})}{p_x},\, e^*\right) \;-\; p_x \;>\; 0 \tag{A.13}$$

Since $f_{xx} < 0$, f_x is strictly decreasing in x^*. A household with lower κ_i has lower x^*, a higher marginal product of inputs $f_x(x^*)$, and therefore a higher shadow price μ_i. Formally:

$$\frac{\partial \mu_i}{\partial \kappa_i} \;=\; A_{idt}\, f_{xx}(x^*, e^*) \cdot \frac{1}{p_x} \;<\; 0 \tag{A.14}$$

Financially poorer households face a higher shadow cost of the constraint and thus have more to gain from its relaxation through aid disbursements.

A.5 Comparative Statics: Proof of Propositions 1 and 2

Financial Channel (Proposition 1). Equation (A.12) depends on D_{dt} through $x^* = (\kappa_i + B(D_{dt}))/p_x$. Differentiating with respect to D_{dt} and invoking the envelope theorem for e^* (since e^* satisfies Equation (A.11), its marginal contribution to the objective is zero):

$$\left.\frac{\partial Y^*_{idt}}{\partial D_{dt}}\right|_{\text{financial}} \;=\; \frac{\partial\left[A_{idt}\, f(x^*, e^*)\right]}{\partial x^*} \cdot \frac{\partial x^*}{\partial D_{dt}} \;=\; A_{idt}\, f_x(x^*, e^*) \cdot \frac{B'(D_{dt})}{p_x} \;>\; 0 \tag{A.15}$$

since $A_{idt} > 0$, $f_x > 0$, and $B' > 0$ by assumption. This establishes the first part of Proposition 1.

For the heterogeneity result, differentiate Equation (A.15) with respect to κ_i, noting that $x^* = (\kappa_i + B)/p_x$ depends on κ_i:

$$\frac{\partial^2 Y^*_{idt}}{\partial D_{dt}\, \partial \kappa_i} \;=\; A_{idt}\, f_{xx}(x^*, e^*) \cdot \frac{\partial x^*}{\partial \kappa_i} \cdot \frac{B'(D_{dt})}{p_x} \;=\; A_{idt}\, f_{xx}(x^*, e^*) \cdot \frac{B'(D_{dt})}{p_x^2} \;<\; 0 \tag{A.16}$$

since $f_{xx} < 0$ and all other terms are positive. This establishes the second part of Proposition 1: the financial-channel yield gain is strictly decreasing in κ_i. □

Knowledge Transfer Channel (Proposition 2). Differentiating Equation (A.12) with respect to A_{idt}, applying the envelope theorem with respect to effort and treating x^* as fixed (since the financial channel is held separately):

$$\frac{\partial Y^*_{idt}}{\partial A_{idt}} \;=\; f(x^*, e^*) \;>\; 0 \tag{A.17}$$

The total derivative of Y^*_{idt} with respect to D_{dt} through the knowledge transfer channel is obtained by the chain rule:

$$\left.\frac{\partial Y^*_{idt}}{\partial D_{dt}}\right|_{\text{info}} = \frac{\partial Y^*_{idt}}{\partial A_{idt}} \cdot \frac{\partial A_{idt}}{\partial D_{dt}} = f(x^*, e^*) \cdot \rho'(D_{dt}) \cdot (A_{\max} - A^0_i) \geq 0 \tag{A.18}$$

where $\partial A_{idt}/\partial D_{dt} = \rho'(D_{dt})(A_{\max} - A^0_i)$ follows from differentiating Equation (7). Equation (A.18) is positive whenever $A^0_i < A_{\max}$ (the household is below the frontier) and $\rho'(D_{dt}) > 0$.

For the heterogeneity result, differentiate Equation (A.18) with respect to baseline knowledge A^0_i:

$$\frac{\partial}{\partial A^0_i}\left[\left.\frac{\partial Y^*_{idt}}{\partial D_{dt}}\right|_{\text{info}}\right] = -\rho'(D_{dt}) \cdot f(x^*, e^*) < 0 \tag{A.19}$$

The knowledge-transfer-channel yield gain is strictly decreasing in A^0_i: households further from the technology frontier benefit more from aid-funded extension. Since formal education is positively correlated with A^0_i across households, less educated farmers gain more from this channel. □

Aggregate Yield Response. $Y^*_{idt} = A_{idt} f(x^*, e^*)$ depends on D_{dt} through two distinct arguments: x^* (via $B(D_{dt})$) and A_{idt} (via $\rho(D_{dt})$). By the chain rule, and invoking the envelope theorem to eliminate the indirect effect through e^*, the total derivative is the sum of the two mechanism-specific derivatives:

$$\frac{\partial Y^*_{idt}}{\partial D_{dt}} = \underbrace{A_{idt} f_x \cdot \frac{B'}{p_x}}_{\geq 0} + \underbrace{\rho'(D_{dt}) \cdot (A_{\max} - A^0_i) \cdot f}_{\geq 0} > 0 \tag{A.20}$$

This corresponds to Equation (9) in the main text.

A.6 Second-Order Conditions

To confirm that the KKT point is a global maximum, it suffices to show that $\mathscr{L}$ is strictly concave in (x, e). The Hessian of $\mathscr{L}$ with respect to (x, e) is:

$$H_{\mathscr{L}} = \begin{pmatrix} A_{idt} f_{xx} & A_{idt} f_{xe} \\ A_{idt} f_{xe} & A_{idt} f_{ee} - c''(e)/\theta_i \end{pmatrix} \tag{A.21}$$

The leading principal minors are:

$$H_1 = A_{idt} f_{xx} < 0 \tag{A.22}$$

$$H_2 = A^2_{idt}\left[f_{xx}\left(f_{ee} - \frac{c''(e)}{A_{idt}\,\theta_i}\right) - f^2_{xe}\right] \tag{A.23}$$

Since $f_{xx} < 0$ and $f_{ee} - c''/(A_{idt}\theta_i) < 0$ (both terms are negative), their product is positive. By the strict concavity of f, $f_{xx} f_{ee} > f^2_{xe}$, which ensures $H_2 > 0$. Therefore $H_{\mathscr{L}}$ is negative definite: the objective is strictly concave in (x, e), the KKT conditions are sufficient, and the solution (x^*, e^*) is a unique global maximum.

B Data Construction

This appendix describes the construction of the three datasets used in the analysis: aid data, household survey data, and contextual indicators. It also documents their harmonization and the construction of the estimation sample.

B.1 Aid Data

Source. Aid data come from the Geocoded Official Development Assistance Dataset (GODAD, Version 1.0), developed by Bomprezzi et al. (2024) and hosted by the University of Göttingen. GODAD combines records from the OECD Creditor Reporting System (CRS) with data on Chinese public finance and World Bank projects. Each project is geocoded at the subnational level (ADM1 and ADM2) and provides the project identifier, title, donor, DAC sector codes, committed and disbursed amounts, start and payment years, and the GADM region identifier. When a project is associated with multiple locations, its amount is distributed evenly across those locations. GODAD also provides aggregated district-by-year panels of commitments and disbursements in constant 2014 USD, disaggregated by donor, flow type, and sector.

Geographic Scope. The analysis retains projects located in the eight WAEMU member countries: Benin, Burkina Faso, Côte d'Ivoire, Guinea-Bissau, Mali, Niger, Senegal, and Togo. All eight countries have household survey data available for both survey rounds and therefore enter the estimation sample. We retain all eight countries throughout the analysis to preserve the national aid portfolios needed to construct the leave-one-out instrument.

Classification of Agricultural Projects. Projects are classified as agricultural using two complementary criteria. A project is classified as agricultural if it satisfies either criterion.

The first criterion is based on DAC sector codes. A project is classified as agricultural if it is assigned at least one of the following codes: 310 (agriculture, forestry, and fishing), 311 (agriculture), 312 (forestry), 313 (fishing), 410 (environmental protection and natural resources), 520 (development food aid and food security), 720 (emergency food aid), or 730 (reconstruction relief).

The second criterion is based on keywords in the project title. The title is searched for the following terms: "agri", "farm", "livestock", "fish", "irrig", "pastoral", "cotton", "rice", "cereal", "food aid", and "food security". Projects satisfying either criterion are classified as agricultural. Because code 410 is relatively broad, we assess the sensitivity of the results to its inclusion. Code 410 accounts for 496 of the 8,149 projects classified as agricultural, or approximately 6 percent. Excluding these projects and re-estimating the preferred specification leaves the coefficient essentially unchanged ($\hat{\beta} = 0.126$ versus 0.127), while the Kleibergen–Paap F-statistic increases from 18.49 to 20.99.

Temporal Reference for Commitments and Disbursements. Commitments and disbursements are assigned different reference years. For commitments, defined as amounts formally approved, the reference year is the project start year. For disbursements, defined as funds actually transferred, the reference year is the payment year when available and the start year otherwise. This distinction is consequential because commitments may occur at project initiation whereas disbursements can extend over subsequent years, generating the difference between commitment and disbursement coverage documented in Table B.1.

Aggregation to the District-by-Year Level. Aid flows are aggregated to the country-by-district-by-year level, separately by donor, flow type, and sector. Six aggregates are constructed: (i) total Chinese commitments; (ii) Chinese agricultural commitments; (iii) total World Bank commitments; (iv) World Bank agricultural commitments; (v) total World Bank disbursements; and (vi) World Bank agricultural disbursements. For each aggregate, we also compute the number of distinct active projects in each district-year. When a project spans multiple locations within the same district, it is counted only once to avoid double-counting.

Integration with the District Panel and Treatment of Zeros. The six aid aggregates are merged into a comprehensive district panel covering all eight WAEMU countries. District-years without an active agricultural project are assigned a disbursement value of zero rather than treated as missing, so that the absence of aid is explicitly represented in the data.

Treatment Variable. The primary treatment variable is World Bank agricultural disbursements at the district-year level, expressed in constant USD and log-transformed as described in the main text. Commitment amounts are constructed analogously and used in alternative specifications. Binary indicators for positive disbursements and counts of active projects are also constructed for robustness analyses.

Sample Restriction. The full aid panel spans the years available in GODAD. We restrict the panel to 2018 and 2021 to match the two household-survey waves. The resulting panel contains 686 district-year observations, corresponding to 343 ADM2 districts observed in both years. The districts are distributed across the eight WAEMU countries as follows: Benin (76), Burkina Faso (45), Côte d'Ivoire (33), Guinea-Bissau (37), Mali (50), Niger (36), Senegal (45), and Togo (21).

B.2 Household Survey Data

Source and Waves. Household data come from the two waves of the Harmonized Survey on Household Living Conditions (EHCVM), implemented by national statistical offices in collaboration with the World Bank and the WAEMU Commission. The first wave covers the 2018–2019 survey period and the second covers 2021–2022. The 2021–2022 survey includes a longitudinal component that follows a subset of households interviewed in 2018–2019. We use the corresponding panel observations for the household fixed-effects analysis. All eight WAEMU countries have observations in both waves and enter the estimation sample.

Outcome Variables. The EHCVM contains detailed plot-level information on crop production, cultivated area, labor use, and agricultural inputs, allowing us to construct several measures of farm performance. Unless otherwise noted, continuous outcome variables are expressed in logarithms.

- *Crop yield (kg/ha).* Our primary outcome is household crop yield, measured as total crop production in kilograms divided by total cultivated area in hectares, aggregated across all plots and crops.
- *Net agricultural income.* Household net revenue from crop and livestock production after deducting input expenditures. Results are reported in Table 3.
- *Labor productivity.* Agricultural output value per day of family agricultural labor. Results are reported in Table 3.

- *Value of output per hectare.* Total crop production valued at local market prices, divided by total cultivated area. Results are reported in Table 3.
- *Cereal yield.* Crop yield computed using cereal crops only, including rice, millet, sorghum, and maize. Results are reported in Table 5.
- *Non-cereal yield.* Crop yield computed using non-cereal crops, including groundnut, cowpea, okra, sorrel, and watermelon. Results are reported in Table 5.
- *Food-crop yield.* Crop yield computed over all food crops, combining cereals and non-cereal food crops. Results are reported in Table 5.
- *Cash-crop yield.* Crop yield computed for cash crops, which in our sample consist primarily of groundnut and are mutually exclusive from the cereal category. Results are reported in Table 5.

Baseline Control Variables. The following variables enter every regression specification as controls. Cross-country harmonization is described below where applicable.

- *Household size.* Total number of household members.
- *Head's age.* Age of the household head in years.
- *Head's age*^{2}. Squared age of the household head, included to capture nonlinear age effects.
- *Male head (0/1).* Binary indicator equal to one if the household head is male. The variable is harmonized across the French- and Portuguese-language survey instruments.
- *Secondary education (0/1).* Binary indicator equal to one if the household head has attained at least lower-secondary education. Education is standardized into a nine-level scale: 0 (none), 1 (preschool), 2–4 (primary cycles 1–3), 5–6 (lower and upper secondary), 7 (technical/vocational), 8 (post-secondary), and 9 (tertiary). The indicator equals one for level 5 or above.
- *Cultivated area (log ha).* Log of total cultivated area in hectares, aggregated across all plots operated by the household.
- *Chemical fertilizer (0/1).* Binary indicator equal to one if the household uses chemical fertilizer on at least one plot.
- *Motorized tillage (0/1).* Binary indicator equal to one if the household uses mechanical traction, such as a tractor or power tiller, on at least one plot. Missing or empty responses are coded as zero.
- *Distance to city (km).* Distance in kilometers from the household's enumeration area to the nearest city, retained in levels.
- *Religion.* Categorical indicators for the religion of the household head, with the modal category omitted as the reference category.
- *Urban (0/1).* Binary indicator equal to one if the household resides in an urban area, harmonized across country-specific classifications of residential milieu.
- *District Gini.* Gini coefficient of per capita expenditure computed from the EHCVM consumption module at the ADM2 district level, capturing local inequality as a contextual control.

Additional Controls Used in Robustness and Heterogeneity Specifications.

- *Number of plots.* Total number of cultivated plots operated by the household.
- *Intercropping (0/1).* Binary indicator equal to one if the household practices intercropping on at least one plot.
- *Improved seeds (0/1).* Binary indicator equal to one if the household uses improved or certified seeds on at least one plot.

- *ICT access (0/1).* Binary indicator equal to one if the household owns or regularly accesses a mobile phone or radio.
- *Media access (0/1).* Binary indicator equal to one if the household has access to media, including television, radio, or newspapers.
- *Electricity access (0/1).* Binary indicator equal to one if the household is connected to the electricity grid. Harmonized from French and Portuguese survey responses.
- *Civilian attacks.* Number of violent events targeting civilians in the district-year, obtained from ACLED. See Appendix B.3.

Panel Structure. A unique household identifier is constructed by interacting the survey household identifier with the country code. The data are organized at the household-by-year level. The longitudinal component of the EHCVM follows only a subset of households across the two waves, so the resulting panel is unbalanced. The household fixed-effects estimator therefore uses households observed in both 2018–2019 and 2021–2022.

B.3 Conflict

Armed conflict data come from the Armed Conflict Location and Event Data Project (ACLED) (Raleigh et al., 2010), which provides georeferenced conflict events with detailed temporal and spatial information. We aggregate events targeting civilians to the district-by-year level and construct a binary indicator equal to one when at least one such event occurs in the district-year. The resulting measures are used in the conflict heterogeneity analysis.

B.4 Data Sources and Variables for Mechanism Validation

The mechanism analysis in Tables D.7–D.9 uses intermediate outcomes drawn from two modules of the Harmonized Survey on Household Living Conditions (EHCVM): the household questionnaire and the community questionnaire. The community questionnaire is administered at the village or neighbourhood level through interviews with local informants and collects information on local infrastructure, institutions, and agricultural conditions. Community-level variables are merged with the household data using the enumeration area (*grappe*) and survey year.

Financial Constraints and Input Use (Table D.7). The variables in this group are drawn from the EHCVM household questionnaire.

- *Any agricultural input purchased (0/1).* Binary indicator equal to one if the household purchased at least one agricultural input, including seeds, fertilizer, or pesticides, during the agricultural season.
- *Total agricultural labor days.* Total number of days of agricultural labor mobilized by the household, combining family and hired labor across plots and crops during the agricultural season.
- *Agricultural credit or own funds (0/1).* Binary indicator equal to one if the household reports financing agricultural activities through formal or informal credit or through its own funds.

Technology Adoption and Information Access (Table D.8). The first three variables are drawn from the EHCVM household questionnaire, while the remaining variables are drawn from Section S03 of the community questionnaire.

Household questionnaire:

- *Crop diversification: Simpson index.* Household-level Simpson diversity index computed from crop-area shares across all plots:

$$1-\sum_k s_k^2,$$

where s_k denotes the share of cultivated area devoted to crop k. Higher values indicate greater diversification.
- *ICT access (0/1).* Binary indicator equal to one if the household has access to information and communication technologies, including telephone or internet services.
- *Media access: radio or television (0/1).* Binary indicator equal to one if the household has access to a radio or television.

Community questionnaire (Section S03 EHCVM):

- *Mechanised farming in community (0/1).* Binary indicator equal to one if mechanised farming using tractors, power tillers, or other motorised equipment is reported in the community.
- *Improved seeds used in community (0/1).* Binary indicator equal to one if certified or improved seed varieties are reported as being used in the community.
- *Community telephone network coverage (0/1).* Binary indicator equal to one if the community is covered by a mobile telephone network.

Market Access and Institutional Development (Table D.9). All five variables are drawn from the EHCVM community questionnaire and are binary indicators. The corresponding questionnaire items are reported below.

- *Active land rental market in community (0/1).* Binary indicator equal to one if land can be rented or leased in the community (question s03q22, Section S03 EHCVM).
- *Commercialization cooperative in community (0/1).* Binary indicator equal to one if a cooperative specializing in the commercialization of agricultural produce is present in the community (question s03q04_2, Section S03 EHCVM).
- *Agricultural cooperative in community (0/1).* Binary indicator equal to one if a general agricultural cooperative is present in the community (question s03q03, Section S03 EHCVM).
- *Input-supply cooperative in community (0/1).* Binary indicator equal to one if a cooperative specializing in the supply of agricultural inputs is present in the community (question s03q04_1, Section S03 EHCVM).
- *Agricultural cooperative: infrastructure source (0/1).* Binary indicator equal to one if an agricultural cooperative is identified as a source of agricultural infrastructure or services in the community (question s01q14_5, Section S01 EHCVM).

B.5 Panel Construction and Estimation Sample

The estimation dataset is constructed by combining four components. Household records from the EHCVM form the base dataset. District-level World Bank aid data are merged at the district-by-year level, while conflict measures from ACLED are merged using the same geographic and temporal identifiers. A district-level Gini coefficient of per capita expenditure is calculated from the EHCVM consumption module and merged at the district level.

Table B.1: Descriptive Statistics

Variable	Mean	Std. Dev.	Min	Max
Outcome				
Log yield (kg/ha)	6.239	1.785	−2.731	16.604
Treatment				
Log WB agri. disbursements	8.628	6.361	0.000	18.318
Log WB agri. commitments	3.185	6.363	0.000	17.124
WB agri. disb. >0 (0/1)	0.655		0	1
Household controls (baseline)				
Household size	7.14	4.39	1	56
Age of HH head	47.77	14.31	15	105
Head's age^2	2,484	1,421	225	11,025
Male HH head (0/1)	0.881		0	1
Secondary education (0/1)	0.036		0	1
Log cultivated area (ha)	0.585	1.467	−5.298	4.501
Chemical fertilizer use (0/1)	0.414		0	1
Motorized tillage (0/1)	0.036		0	1
Distance to city (km)	22.64	23.71	0	200
Urban (0/1)	0.209		0	1
District Gini	0.575	0.036	0.526	0.642
Additional controls (robustness specifications)				
Number of plots	2.272	1.518	0	20
Intercropping (0/1)	0.389		0	1
Improved seeds (0/1)	0.160		0	1
ICT access (0/1)	0.887		0	1
Media access (0/1)	0.482		0	1
Electricity grid access (0/1)	0.372		0	1
Contextual (district-year)				
Civilian attacks (count)	2.083	7.617	0	76

Notes: This table presents summary statistics for all variables used in the analysis. $N = 36{,}673$ household-year observations in the full merged dataset (before dropping observations with missing farm controls). The regression sample is approximately 28,726 observations. Binary variables (0/1) are shown as proportions; standard deviations are omitted for these. Civilian attacks is the district-year count of violent events targeting civilians (ACLED).

The merged dataset contains 36,673 household-year observations. We then restrict the sample to observations with non-missing values for the log yield outcome and the household, farm, and district controls used in the baseline specification. This yields 28,726 household-year observations across 308 districts in the eight countries and the two survey rounds, 2018 and 2021. The district-level treatment is assigned to each household according to the district and survey year in which the household is observed. District-year observations without World Bank agricultural disbursements are assigned a value of zero before the logarithmic transformation.

Table B.1 reports summary statistics for the outcome, treatment, control, and contextual variables used throughout the analysis.

B.6 Panel Attrition

Household fixed effects are identified only for households observed in both survey waves. Of the 23,554 households observed in 2018, 21,416 (91%) are re-interviewed in 2021, implying an attrition rate of 9%. Selective attrition could bias the within-household estimates if the probability of remaining in the panel is systematically related to aid exposure. Table B.2 examines this possibility by regressing an indicator for reappearing in the 2021 wave on 2018 district-level log disbursements, using the 2018 household sample. Column (1) shows a small but statistically significant unconditional association between baseline aid exposure and subsequent re-interview status. Column (2) adds the full baseline control vector and country fixed effects, closely matching the main specification within a single cross-section. The coefficient declines substantially in magnitude and is no longer statistically significant, suggesting that the unconditional association reflects observable differences across households and countries rather than a systematic relationship between aid exposure and panel retention. Attrition is similarly unrelated to the leave-one-out instrument conditional on the same controls. Overall, these results provide little evidence that selective attrition is an important source of bias in the household fixed-effects estimates.

Table B.2: Panel Attrition: Reappearance in 2021 on 2018 Aid Exposure

	(1) No controls	(2) Controls + Country FE
$\text{Aid}_{d,2018}$	-0.004***	-0.001
	(0.001)	(0.001)
Controls		✓
Country FE		✓
Observations	23,554	19,886

Notes: This table presents linear probability models of an indicator for reappearing in the 2021 wave, estimated on the 2018 baseline household sample. Column (1) is unconditional. Column (2) adds the full baseline control vector and country fixed effects. Standard errors clustered at ADM2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B.7 Identifying Variation in the Leave-One-Out Instrument

The leave-one-out (LOO) instrument is constructed at the district $\times$ year level. For each district d in country c and year t, the instrument is the mean of log agricultural disbursements across all other districts in the same country-year:

$$Z_{dt} = \frac{1}{|\mathscr{D}_c| - 1} \sum_{\substack{d' \in \mathscr{D}_c \\ d' \neq d}} \text{Aid}_{d't} \tag{B.24}$$

where $\text{Aid}_{d't}$ denotes log agricultural disbursements in district d' at time t (as defined in Section 3.2) and $\mathscr{D}_c$ denotes the set of districts in country c. The LOO mean is computed using the district-level aid data before merging with the household survey. The instrument is therefore defined at the district level and does not mechanically incorporate the focal district's own disbursements. It is subsequently merged into the household panel alongside the treatment variable.

Section 3.3 argues that instrument relevance arises from country-level aid programming cycles that generate common movements in agricultural disbursements across districts within a country and year. Table B.3 and Figure B.1 characterize the variation underlying this mechanism directly. Decomposing the variance of Z_{dt} into between- and within-country-year components shows that 99.94 percent of its total variation occurs between country-year cells, while only 0.06 percent occurs within them. The LOO instrument is therefore predominantly a country-year-level measure of aid exposure, with only limited district-level dispersion around the country-year mean.

The concentration of variation across country-year cells follows directly from the LOO construction. For two districts d and d' in the same country-year,

$$Z_{dt} - Z_{d't} \;=\; \frac{\text{Aid}_{d't} - \text{Aid}_{dt}}{N_c - 1}, \tag{B.25}$$

where $N_c = |\mathscr{D}_c|$. This is an exact algebraic consequence of the leave-one-out construction, independent of any distributional assumption: within a country-year, the dispersion of the LOO instrument is mechanically attenuated as the number of districts increases.

If district-level disbursements within a country-year were approximately independent, the standard deviation of the LOO mean across districts would satisfy

$$\text{SD}_d(Z_{dt}) \;\approx\; \frac{\text{SD}_d(\text{Aid}_{dt})}{\sqrt{N_c - 1}}, \tag{B.26}$$

where the standard deviations are calculated across districts within the country-year. Unlike the identity above, this relationship is only an approximation: it treats each LOO mean as if computed from independent draws, whereas the LOO means for different districts in the same country-year are in fact constructed from heavily overlapping sets of the same underlying disbursements, so the actual within-cell dispersion need not match this benchmark exactly.

Table B.3 reports, for each of the sixteen country-year cells in the sample, the number of districts, the standard deviation of raw log disbursements, the observed within-cell standard deviation of Z_{dt}, and the standard deviation implied by the approximation above. Figure B.1 plots within-cell dispersion of the instrument against N_c. The correlation is -0.57, while a regression of log within-cell standard deviation on $\log(N_c - 1)$ yields a slope of -0.79 (s.e. 0.29, $p = 0.018$), reasonably close to the -0.5 benchmark implied by treating district-level disbursements as approximately independent, although the limited number of country-year cells and differences in the dispersion of underlying aid across cells naturally generate deviations from this benchmark. The approximation also explains 87 percent of the cross-cell variation in observed within-cell instrument dispersion ($R^2 = 0.868$).

These results clarify the source of identifying variation in the empirical specification. Because the LOO instrument is overwhelmingly determined by country-year aid cycles, a specification saturated with country$\times$year fixed effects would absorb its dominant component by construction. The remaining within-country-year variation is mechanically small and cannot provide the principal source of identification. The main specification therefore uses household and year fixed effects, which retain the cross-country and over-time variation in the LOO instrument while controlling for time-invariant household heterogeneity and common year shocks. This feature also makes the exclusion restriction substantive: identification relies on country-year movements in aid allocations being related to household agricultural outcomes through district aid exposure rather than through unobserved country-year shocks that independently affect those outcomes.

Table B.3: Identifying Variation in the LOO Instrument, by Country-Year Cell

Country	Year	N_c	SD(Aid_{dt})	SD(Z_{dt})	Predicted SD
Benin	2018	74	5.561	0.076	0.651
Benin	2021	74	6.541	0.090	0.766
Burkina Faso	2018	45	3.990	0.091	0.602
Burkina Faso	2021	45	4.100	0.093	0.618
Côte d'Ivoire	2018	33	4.159	0.130	0.735
Côte d'Ivoire	2021	33	3.578	0.112	0.633
Guinea-Bissau	2018	27	5.666	0.218	1.111
Guinea-Bissau	2021	27	3.026	0.116	0.593
Mali	2018	49	5.184	0.108	0.748
Mali	2021	50	4.338	0.089	0.620
Niger	2018	35	2.579	0.076	0.442
Niger	2021	35	1.232	0.036	0.211
Senegal	2018	43	5.696	0.136	0.879
Senegal	2021	43	2.291	0.055	0.354
Togo	2018	21	5.666	0.283	1.267
Togo	2021	21	4.398	0.220	0.983

Notes: This table decomposes the identifying variation in the leave-one-out instrument Z_{dt} by country-year cell. N_c is the number of districts in the cell. SD(Aid_{dt}) is the standard deviation of raw log disbursements across districts in the cell. SD(Z_{dt}) is the actual within-cell standard deviation of the instrument. Predicted SD is SD(Aid_{dt})$/\sqrt{N_c - 1}$, the value implied by treating district-level disbursements as approximately independent draws.

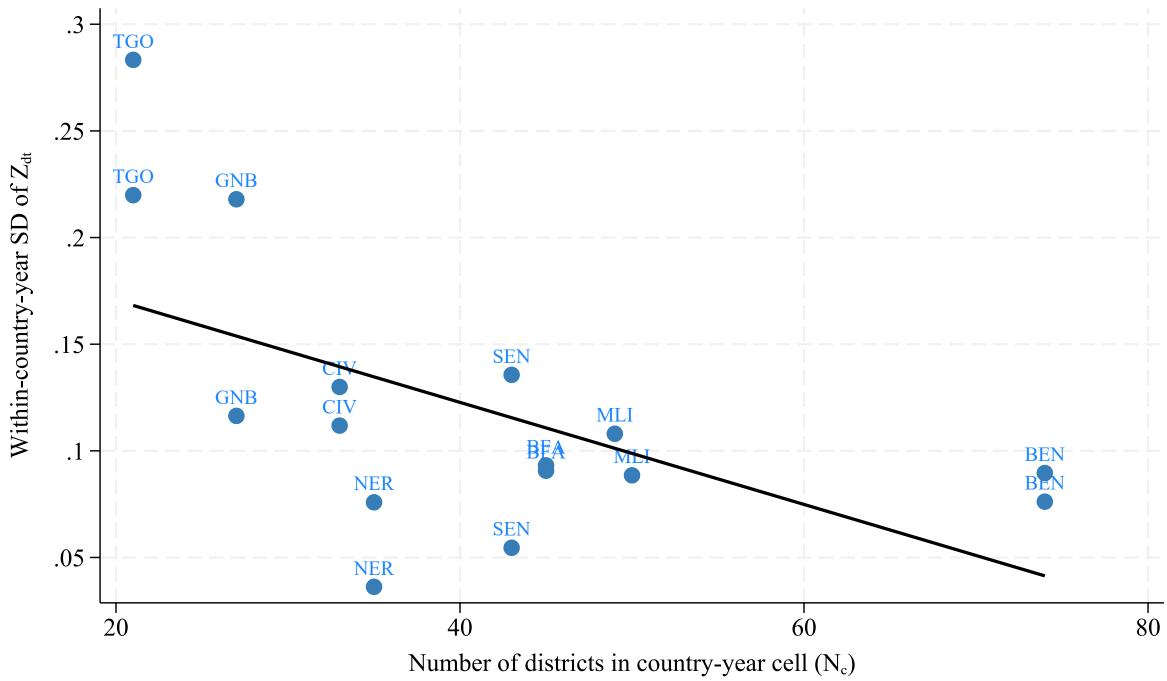

Figure B.1: Within-Country-Year Dispersion of the LOO Instrument, by Number of Districts

Notes: This figure plots the within-country-year standard deviation of the leave-one-out instrument Z_{dt} against the number of districts N_c in the country-year cell, with country-year labels and a fitted line. Sixteen country-year cells (eight countries, two survey waves).

C Additional Robustness and Heterogeneity

This appendix reports a comprehensive set of additional analyses addressing the main identification concerns and heterogeneity questions raised in the paper. Each block is motivated by a specific weakness or open question; the results inform which findings are robust enough to include in the main text.

C.1 COVID-19 Robustness: Locality and Interacted Fixed Effects

The 2021 EHCVM survey wave was conducted during the COVID-19 pandemic and its recovery period. Although year fixed effects absorb shocks common to all countries in a given survey wave, they do not account for heterogeneous country- or locality-level disruptions to agricultural production, input markets, or local economic activity. We therefore examine whether the baseline IV estimate is sensitive to increasingly demanding fixed-effects structures that absorb time-invariant locality characteristics and, subsequently, locality-specific shocks over time.

Column (2) adds locality fixed effects to the preferred baseline specification. These effects absorb time-invariant characteristics of the village or enumeration area, such as local infrastructure, agroecological conditions, and persistent differences in market access. Column (3) replaces the separate year and locality fixed effects with locality$\times$year fixed effects. This specification allows each locality to experience its own year-specific shock and therefore absorbs differential changes in local conditions between survey waves, including locality-specific disruptions associated with the pandemic.

If the estimated coefficient remains stable when these increasingly flexible fixed-effects structures are introduced, the result is less likely to be driven by unobserved time-invariant locality characteristics or by locality-specific changes between survey waves.

Table C.4: COVID-19 Robustness

	(1) Baseline	(2) +Locality	(3) +Loc$\times$yr
Aid$_{dt}$	0.127***	0.127***	0.188***
	(0.042)	(0.042)	(0.067)
Controls	✓	✓	✓
Household FE	✓	✓	✓
Country FE	✓	✓	✓
Year FE	✓	✓	
Locality FE		✓	
Loc. $\times$ Year FE			✓
Observations	28,726	28,726	28,640
KP F-stat	18.48	18.48	12.62

Notes: This table shows the stability of the baseline IV estimate to progressively demanding fixed-effects structures. Column (1) replicates the preferred baseline specification (column 3 of Table 1), with the full baseline control vector and household and year fixed effects; column (2) adds locality fixed effects on top of the baseline; column (3) replaces year and locality fixed effects with locality$\times$year fixed effects, absorbing differential COVID-19 shocks by locality. IV-LOO throughout. Standard errors clustered at ADM2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.2 Additional Robustness Checks

Table C.5 reports three additional robustness checks that address alternative functional-form choices and the potential influence of extreme observations.

Table C.5: Additional Robustness: Transformation and Trimming

	(1) Baseline	(2) asinh	(3) Trimmed
Aid_{dt} (log)	0.127^{***}		0.096^{***}
	(0.042)		(0.035)
Aid_{dt} (asinh)		0.120^{***}	
		(0.038)	
Specification	Preferred specification	asinh transformation	Trimmed 1%–99%
Controls	✓	✓	✓
Household FE	✓	✓	✓
Year FE	✓	✓	✓
Observations	28,726	28,726	27,842
KP F-stat	18.48	17.63	19.26

Notes: This table presents robustness checks for the baseline IV estimate. Column (1): baseline log transformation. Column (2): inverse hyperbolic sine (asinh) transformation (Bellemare and Wichman, 2020). Column (3): sample trimmed at the 1st and 99th percentiles of log yield. IV-LOO with household and year fixed effects (baseline IV, column 3 of Table 1). Standard errors clustered at ADM2. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Column 2 replaces the baseline $\log(\text{Aid}+1)$ transformation with the inverse hyperbolic sine transformation, $\text{asinh}(\text{Aid})$. The baseline transformation accommodates zero disbursements but compresses differences among observations with zero or relatively small aid. The inverse hyperbolic sine provides an alternative transformation that is defined at zero and converges to $\log(2\text{Aid})$ for large positive aid (Bellemare and Wichman, 2020). Similar estimates under the two transformations would indicate that the main result is not sensitive to the treatment of zero and small disbursement values.

Column 3 trims the sample by excluding households in the top and bottom 1% of the log yield distribution. This check assesses whether the estimated effect is disproportionately influenced by observations in the extreme tails of the outcome distribution. The trimmed-sample estimate provides a conservative test of whether the main finding reflects the central mass of the yield distribution rather than a small number of unusually high- or low-yield observations.

C.3 Permutation Placebo Tests

We assess whether the headline associations are unusually large relative to coefficients generated by random reassignment of the treatment and instrument. We conduct two randomization-inference placebo tests, each based on 500 random reassignments. The first targets the OLS treatment association, while the second targets the predictive content of the LOO instrument in the reduced form.

In the first test, we randomly reassign district-level log disbursements across districts within each country-year cell and re-estimate the OLS specification. This procedure preserves the country-year distribution of aid and its temporal structure while breaking the observed correspondence between district-level aid and district-level outcomes. In the second test, we randomly reassign the LOO instrument across all district-year observations and re-estimate the reduced-form regression of yield on the instrument. Because the LOO instrument is predominantly determined by country-year aid cycles, a permutation restricted within country-year cells would leave most of the instrument unchanged. We therefore permute the instrument across the full sample, thereby breaking its observed correspondence with the outcome while preserving its empirical marginal distribution.

For each test, the randomization p-value is calculated as

$$p_{\text{perm}} = \frac{1}{R}\sum_{r=1}^{R}\mathbf{1}_{\left\{\left|\hat{\beta}^{(r)}_{\text{placebo}}\right|\geq\left|\hat{\beta}_{\text{true}}\right|\right\}}, \tag{C.27}$$

where $R = 500$ denotes the number of random reassignments. The observed coefficient is therefore evaluated against the empirical distribution generated under the corresponding randomization scheme.

Both observed coefficients lie in the tails of their respective placebo distributions. The true OLS coefficient is 0.021, and only 9 of the 500 random reassignments produce an absolute coefficient at least as large, yielding a permutation p-value of 0.018. The placebo coefficients are centered close to zero, with a mean of 0.009. The corresponding reduced-form coefficient is 0.130; none of the 500 random reassignments produces an absolute coefficient as large as the observed estimate, implying $p_{\text{perm}} < 0.002$. The reduced-form placebo distribution is centered essentially at zero, with a mean of 0.000, and 4 of the 500 replications exceed the conventional 5% threshold.

These results indicate that the observed OLS association and the reduced-form relationship between the LOO instrument and yields are both unusually large relative to the distributions generated by the corresponding randomization schemes. The placebo exercises therefore provide evidence against an explanation based solely on random reassignment of district-level aid or the LOO instrument.

Table C.6: Permutation Placebo Tests

	(1) OLS permute Aid_{dt}	(2) IV reduced form permute Z_{dt}
True coefficient	0.021^{**}	0.130^{***}
	(0.009)	(0.030)
Permutation p-value	0.018	<0.002
Placebo mean (SD)	0.009 (0.006)	0.000 (0.007)
Reps. with $\lvert\hat{\beta}_{\text{placebo}}\rvert \geq \lvert\hat{\beta}_{\text{true}}\rvert$	9/500	0/500
Placebos significant at 5%	21/500 (4.2%)	4/500 (0.8%)
Replications	500	500

Notes: This table presents two randomization-inference placebo tests, 500 replications each. Column (1) permutes district-level log disbursements across districts within each country-year cell and re-estimates the OLS yield specification. Column (2) permutes the leave-one-out instrument across all district-years and re-estimates the reduced form of yield on the instrument. "True coefficient" is the estimate on the actual (unpermuted) variable, with its analytic cluster-robust standard error in parentheses; stars are based on the permutation p-value (the share of replications with $\lvert\hat{\beta}_{\text{placebo}}\rvert \geq \lvert\hat{\beta}_{\text{true}}\rvert$). A placebo distribution centered on zero, and a true coefficient in its tail, indicate that the relationship does not arise from spatial clustering or random assignment. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ (permutation).

D Mechanism Validation: Full Estimation Results

This appendix reports the full estimation results underlying the mechanism validation discussed in Section 4.4. The analysis examines fourteen intermediate outcomes organized around the three channels proposed in Section 3.1: financial constraint relaxation, technology adoption and information access, and market access and institutional development. For each outcome, we report OLS and IV-LOO estimates under two fixed-effects structures: district and year fixed effects, and the baseline household, country, and year fixed effects. All specifications include the baseline control vector, and standard errors are clustered at the ADM2 district level.

Financial constraint relaxation. The results for the financial constraint channel indicate a positive relationship between World Bank agricultural disbursements and several measures of agricultural resource use. The probability that a household purchases any agricultural input increases by approximately 0.6 percentage points under OLS and 1.8 percentage points under IV, with the coefficient remaining positive and statistically significant across the four specifications. Total agricultural labor demand also increases, with statistical significance in three of the four specifications; significance is lost in the most demanding IV specification with household fixed effects, where the standard error increases substantially. The probability that agricultural activity is financed through own funds or credit increases by approximately 0.1–0.3 percentage points and remains statistically significant across specifications.

The IV estimates are generally larger than the corresponding OLS estimates. This pattern is consistent with attenuation of the OLS relationship, although the difference between the estimators alone does not identify the source of the discrepancy. Overall, the results provide relatively consistent evidence that greater agricultural disbursements are associated with greater access to agricultural inputs, labor resources, and financing.

Table D.7: Mechanism Validation: Financial Constraint

	District FE		Household FE	
	(1) OLS	(2) IV	(3) OLS	(4) IV
Any agricultural input purchased (0/1)				
Aid_{dt}	0.006^{***}	0.018^{***}	0.004^{**}	0.014^{**}
	(0.002)	(0.005)	(0.002)	(0.007)
Observations	36,966	36,966	28,964	28,964
KP F-stat		35.31		18.11
Total agricultural labour days				
Aid_{dt}	6.368^{***}	34.772^{*}	6.916^{**}	38.881
	(2.252)	(19.649)	(3.009)	(27.979)
Observations	39,621	39,621	32,918	32,918
KP F-stat		35.25		18.16
Agricultural credit or own funds (0/1)				
Aid_{dt}	0.001^{**}	0.002^{***}	0.001^{*}	0.003^{**}
	(0.000)	(0.001)	(0.000)	(0.001)
Observations	39,621	39,621	32,918	32,918
KP F-stat		35.25		18.16

Notes: This table presents OLS and IV-LOO estimates of the effect of World Bank agricultural disbursements on three proxies for financial constraint relaxation: any agricultural input purchased (binary); total agricultural labor days; farm financed via own funds or credit (binary). Columns (1)–(2) absorb district and year fixed effects; columns (3)–(4) absorb household and year fixed effects (baseline). All specifications include the baseline control vector. Standard errors clustered at ADM2 in parentheses. KP F-stat refers to the Kleibergen–Paap first-stage F-statistic. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Technology adoption and information access. The results for technology adoption and information access are more heterogeneous but remain broadly supportive of the proposed channel. Crop diversification, ICT access, and mechanization adoption are positive and statistically significant across all four specifications. For these outcomes, the IV estimates are substantially larger than the corresponding OLS estimates, indicating that the relationship is not driven solely by the variation captured by the OLS specifications.

Evidence for the remaining outcomes is weaker. Media access is statistically significant in three of the four specifications, with significance lost in the most demanding IV specification with household fixed effects. Improved seed use is sensitive to specification and reaches conventional statistical significance in only one of the four specifications. Telephone network coverage is positive and statistically significant under OLS but not under IV, indicating that the OLS association does not survive the IV identification strategy. These heterogeneous results suggest that the evidence for technology and information mechanisms is strongest for diversification, ICT access, and mechanization, while the evidence for seed adoption and telephone coverage is comparatively limited.

Table D.8: Mechanism Validation: Technology Adoption and Information Access

	District FE		Household FE	
	(1) OLS	(2) IV	(3) OLS	(4) IV
Crop diversification (Simpson index)				
Aid_{dt}	0.005^{***}	0.019^{***}	0.004^{*}	0.016^{**}
	(0.002)	(0.005)	(0.002)	(0.006)
Observations	39,621	39,621	32,918	32,918
KP F-stat		35.25		18.16
Access to ICT (0/1)				
Aid_{dt}	0.003^{***}	0.006^{***}	0.004^{**}	0.007^{**}
	(0.001)	(0.002)	(0.002)	(0.003)
Observations	39,575	39,575	32,856	32,856
KP F-stat		35.24		18.15
Media access — radio or television (0/1)				
Aid_{dt}	0.006^{***}	0.008^{*}	0.006^{*}	0.010
	(0.002)	(0.004)	(0.003)	(0.007)
Observations	39,575	39,575	32,856	32,856
KP F-stat		35.24		18.15
Mechanised farming in community (0/1)				
Aid_{dt}	0.009^{***}	0.015^{***}	0.008^{*}	0.016^{**}
	(0.003)	(0.006)	(0.004)	(0.008)
Observations	33,423	33,423	26,736	26,736
KP F-stat		34.46		18.49
Improved seeds used in community (0/1)				
Aid_{dt}	0.007^{*}	0.012	0.006	0.014
	(0.004)	(0.009)	(0.006)	(0.013)
Observations	33,423	33,423	26,736	26,736
KP F-stat		34.46		18.49
Community telephone network coverage (0/1)				
Aid_{dt}	0.008^{***}	-0.000	0.008^{**}	-0.001
	(0.002)	(0.006)	(0.003)	(0.009)
Observations	39,621	39,621	32,918	32,918
KP F-stat		35.25		18.16

Notes: This table presents OLS and IV-LOO estimates of the effect of World Bank agricultural disbursements on six technology adoption and information access outcomes: Simpson crop diversification index; ICT access; media access (radio or television); mechanised farming in community; improved seeds in community; telephone network coverage (all binary except the diversification index). Columns (1)–(2): district and year FE; columns (3)–(4): household, country, and year FE (baseline). All specifications include the baseline control vector. Standard errors clustered at ADM2. KP F-stat refers to the Kleibergen–Paap first-stage F-statistic. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Market access and institutional development. The results for market access and institutional development provide additional evidence of changes in the local agricultural environment associated with greater World Bank agricultural disbursements. The indicator for an active land-rental market is positive and statistically significant across all four specifications. The IV estimates imply an increase of approximately 1.6–1.8 percentage points in the probability that the community has an active land-rental market.

The evidence for cooperative development is more mixed. The indicator for a commercialization cooperative is statistically significant in three of the four specifications, losing significance only in the most demanding IV specification with household fixed effects. The indicators for agricultural cooperatives in general and input-supply cooperatives are less robust, reaching statistical significance in only one specification each. The alternative measure of agricultural cooperative presence based on infrastructure sources is positive throughout but statistically significant in only one of the four specifications, the district fixed-effects IV column.

Table D.9: Mechanism Validation: Market Access and Institutional Development

	District FE		Household FE	
	(1) OLS	(2) IV	(3) OLS	(4) IV
Active land rental market in community (0/1)				
Aid_{dt}	0.011^{***}	0.016^{**}	0.011^{**}	0.018^{*}
	(0.004)	(0.007)	(0.006)	(0.011)
Observations	33,423	33,423	26,736	26,736
KP F-stat		34.46		18.49
Commercialisation cooperative in community (0/1)				
Aid_{dt}	0.007^{***}	0.013^{**}	0.007^{*}	0.012
	(0.003)	(0.006)	(0.004)	(0.009)
Observations	39,621	39,621	32,918	32,918
KP F-stat		35.25		18.16
Agricultural cooperative in community (0/1)				
Aid_{dt}	0.004^{*}	0.008	0.005	0.008
	(0.003)	(0.006)	(0.004)	(0.009)
Observations	33,423	33,423	26,736	26,736
KP F-stat		34.46		18.49
Input-supply cooperative in community (0/1)				
Aid_{dt}	0.005^{*}	0.008	0.006	0.010
	(0.002)	(0.006)	(0.004)	(0.008)
Observations	39,621	39,621	32,918	32,918
KP F-stat		35.25		18.16
Agricultural cooperative — infrastructure source (0/1)				
Aid_{dt}	0.004	0.014^{*}	0.004	0.014
	(0.003)	(0.007)	(0.004)	(0.011)
Observations	39,399	39,399	32,542	32,542
KP F-stat		35.13		18.04

Notes: This table presents OLS and IV-LOO estimates of the effect of World Bank agricultural disbursements on five market access and institutional development outcomes: active land rental market in community; commercialisation cooperative; agricultural cooperative; input-supply cooperative; agricultural cooperative (alternative infrastructure source; all binary). Columns (1)–(2): district and year FE; columns (3)–(4): household, country, and year FE (baseline). All specifications include the baseline control vector. Standard errors clustered at ADM2. KP F-stat refers to the Kleibergen–Paap first-stage F-statistic. ${}^{*}p < 0.10$, ${}^{**}p < 0.05$, ${}^{***}p < 0.01$.

Taken together, the mechanism results provide evidence that the estimated effect of agricultural disbursements is accompanied by changes in several intermediate dimensions of agricultural activity. The most consistent patterns concern input use and financing, crop diversification, ICT access, mechanization, and local land-market activity. Evidence for some of the more specific technology and cooperative measures is less robust across specifications. We therefore interpret these results as supporting evidence for the proposed mechanisms rather than as definitive causal tests of the individual transmission channels.